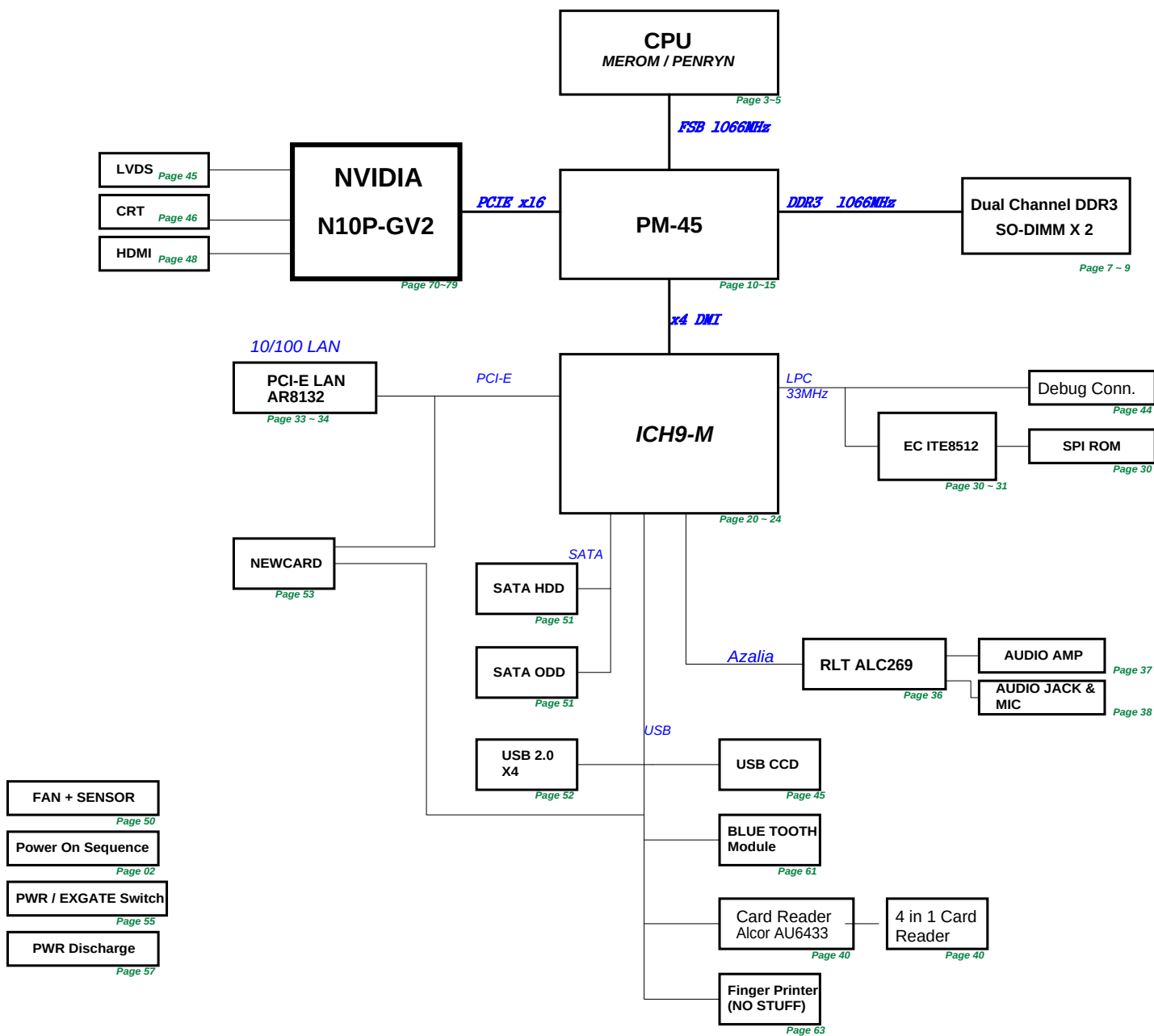
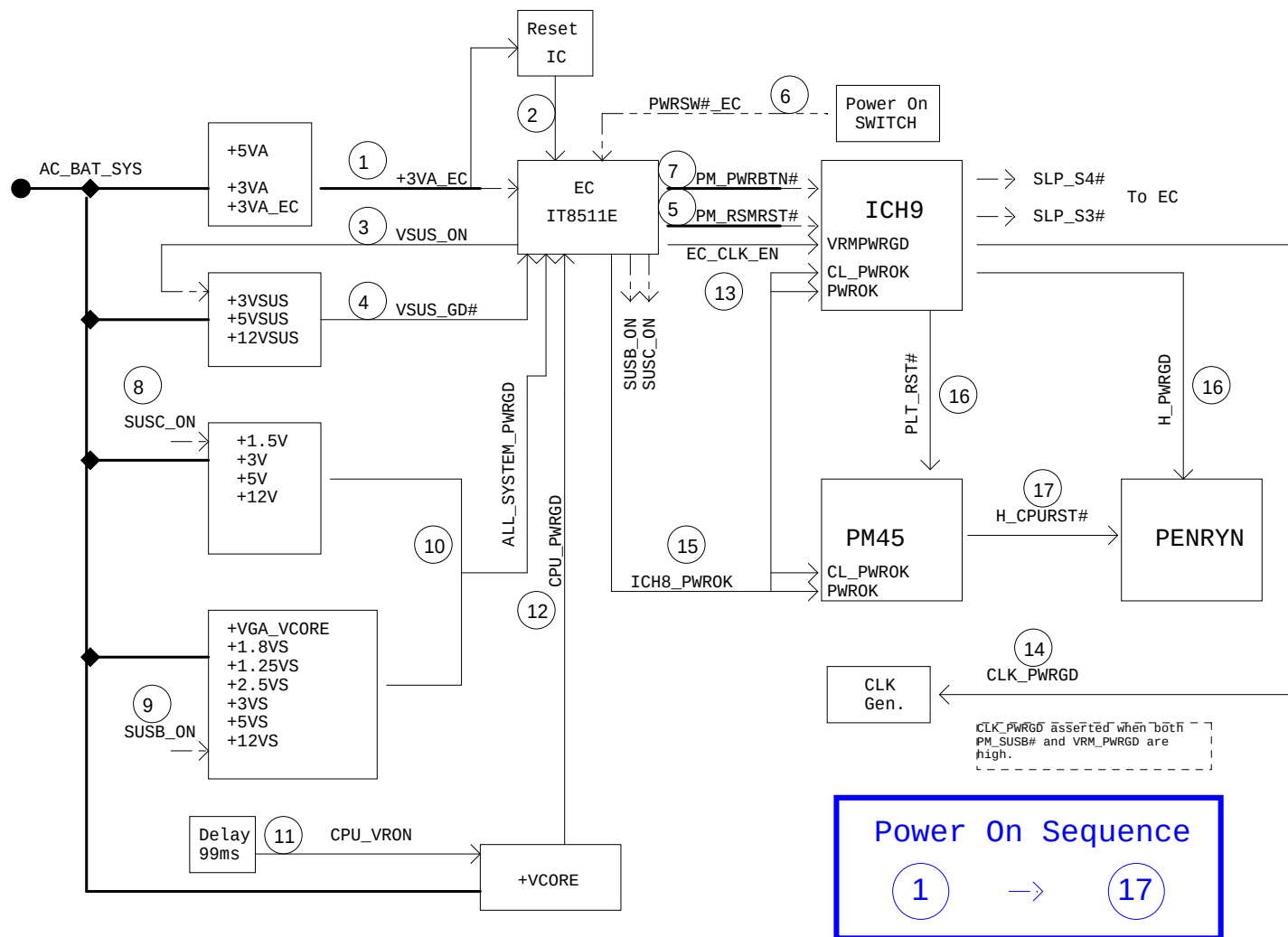
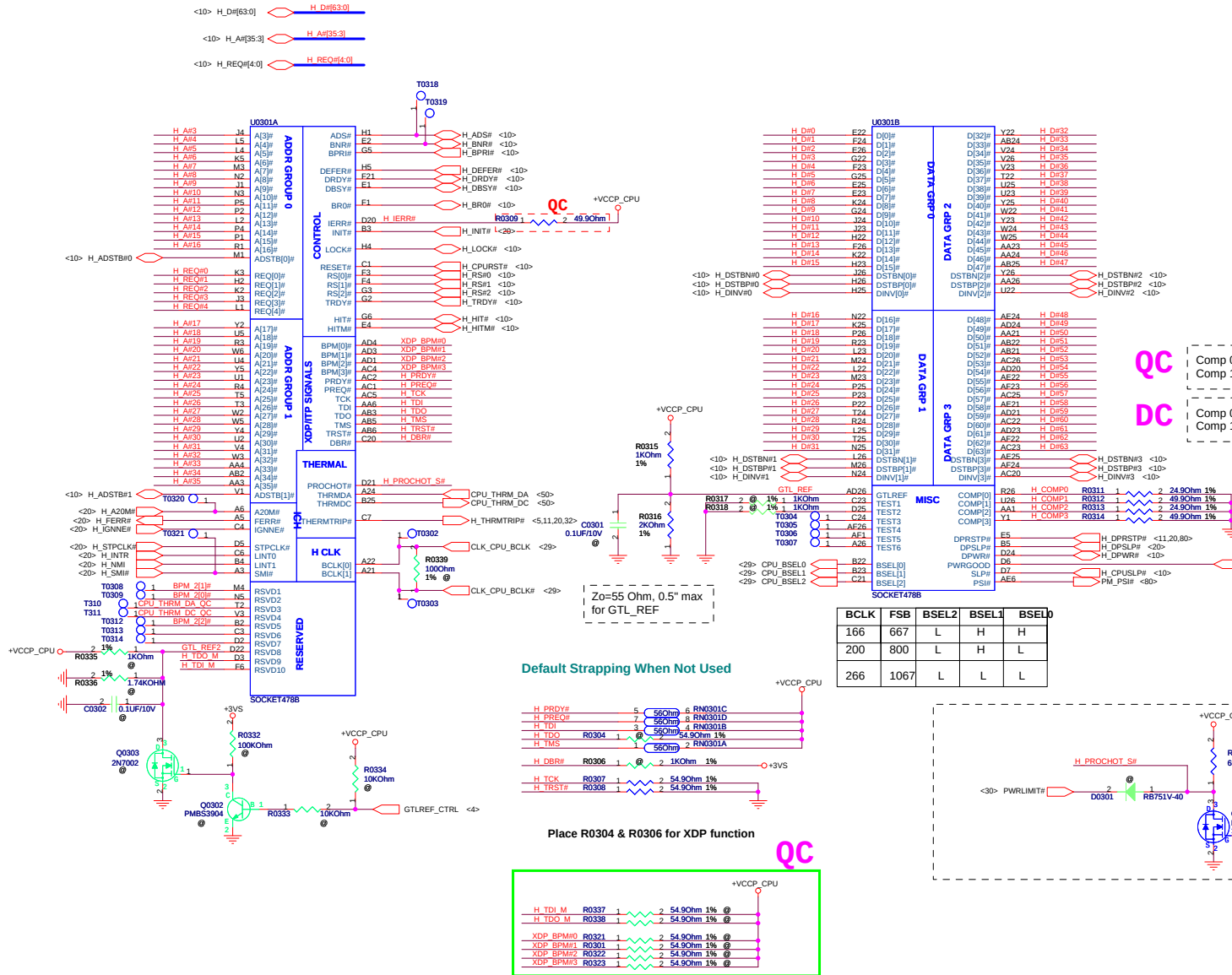


F83Vf MonteVina BLOCK DIAGRAM



Power	
VCORE	Page 80
System	Page 81
1.5VS & 1.05VS	Page 82
DDR & VTT	Page 83
1.8VS	Page 84
VGA_VCORE	Page 85
Charger	Page 88
Detect	Page 90
Load Switch	Page 91
Power Protect	Page 92
Power Protect	Page 93





QC
DC

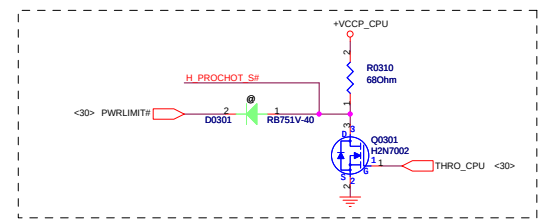
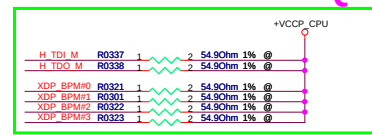
Comp 0,2: Zo=25 Ohm, trace length < 0.5"
Comp 1,3: Zo=50 Ohm, trace length < 0.5"

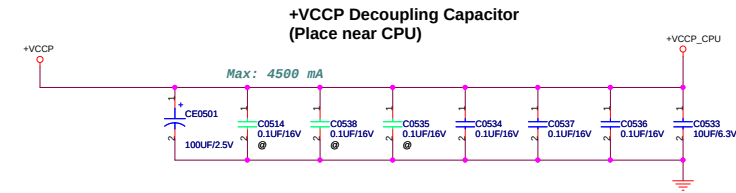
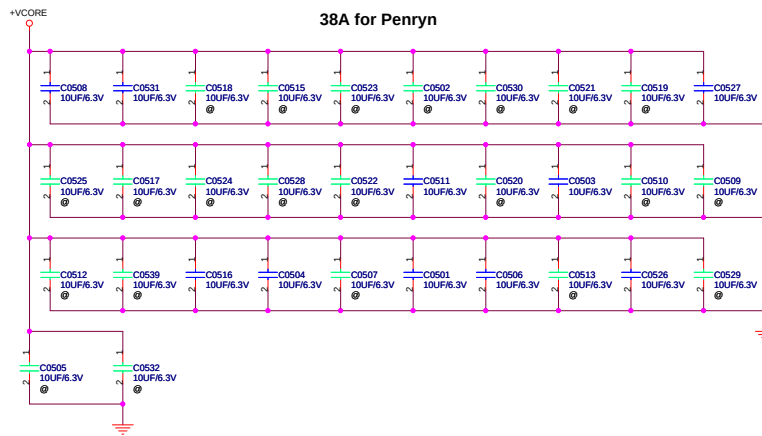
Comp 0,2: Zo=27.4 Ohm, trace length < 0.5"
Comp 1,3: Zo=55 Ohm, trace length < 0.5"

BCLK	FSB	BSEL2	BSEL1	BSEL0
166	667	L	H	H
200	800	L	H	L
266	1067	L	L	L

Default Strapping When Not Used

Place R0304 & R0306 for XDP function



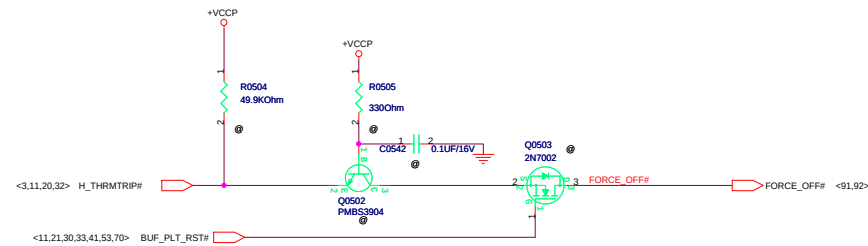


Decoupling guide from Intel

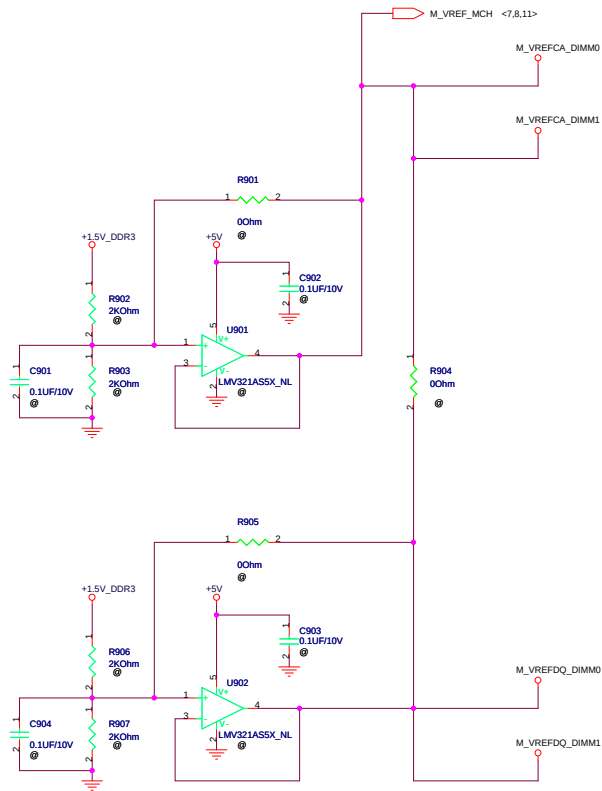
VCCORE	22uF/10V r 10uF	* 32pcs
	330uF/2V	* 6pcs
VCCP	0.1uF	* 6pcs
	150uF	* 1pcs ?
	10uF	* 1pcs ?

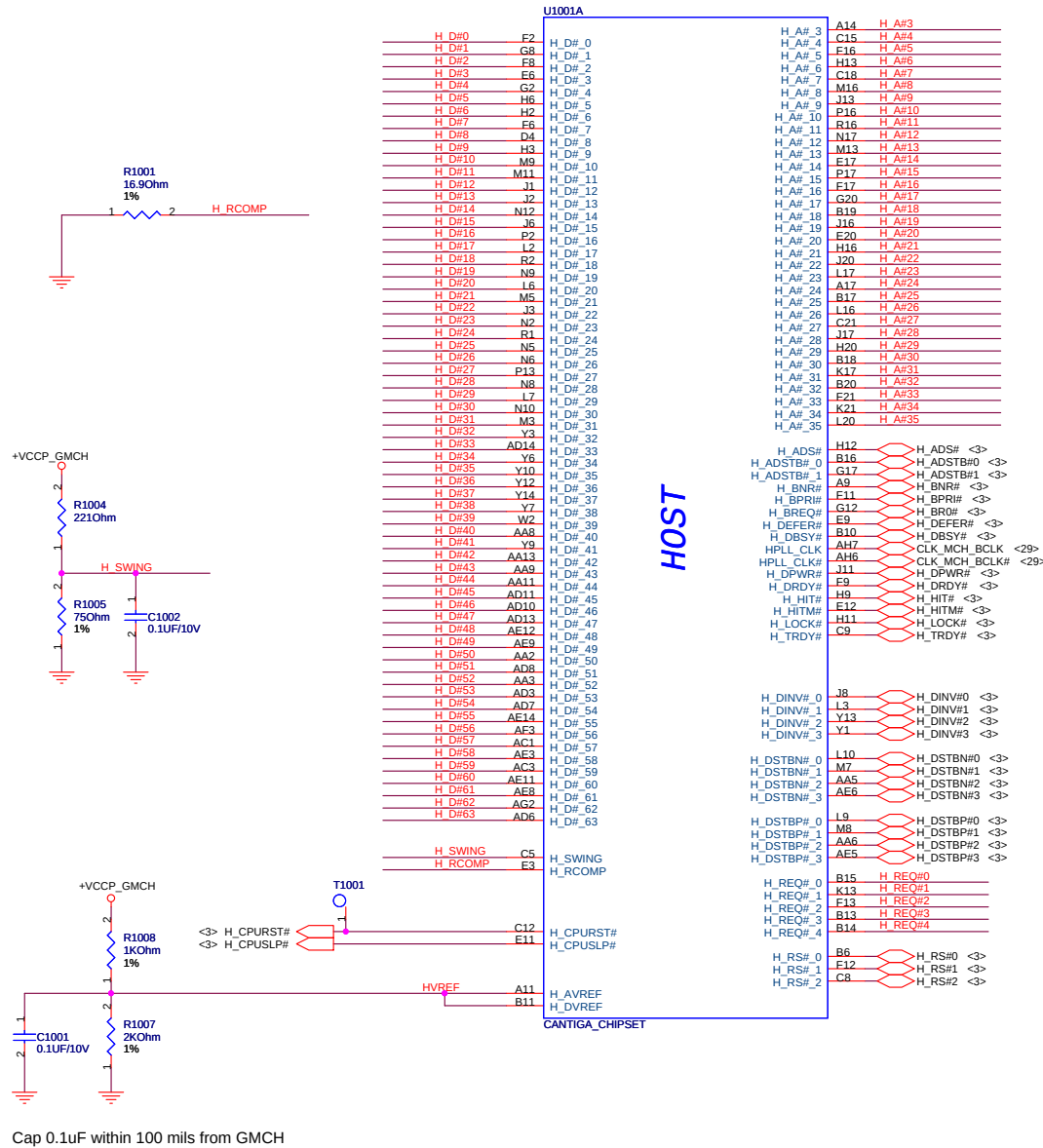
+VCCORE Mid-Frequency Capacitor
Intel: 22uF *32
F83V: 10uF*12

+VCCP Decoupling Capacitor
Intel: 270uF *1, 0.1uF *6
F83: 100uF *1, 0.1uF *3
V1V: ?



Thermal Trip signal (From CPU to ICH-9M and sequence)







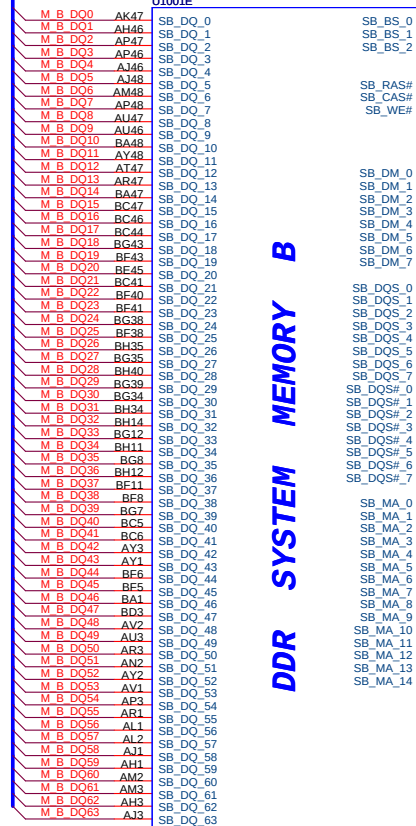
<7> M_A_DQ[0:63]



DDR SYSTEM MEMORY A

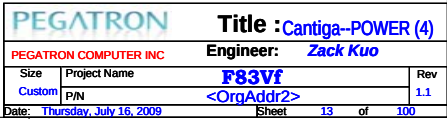
CANTIGA_CHIPSET

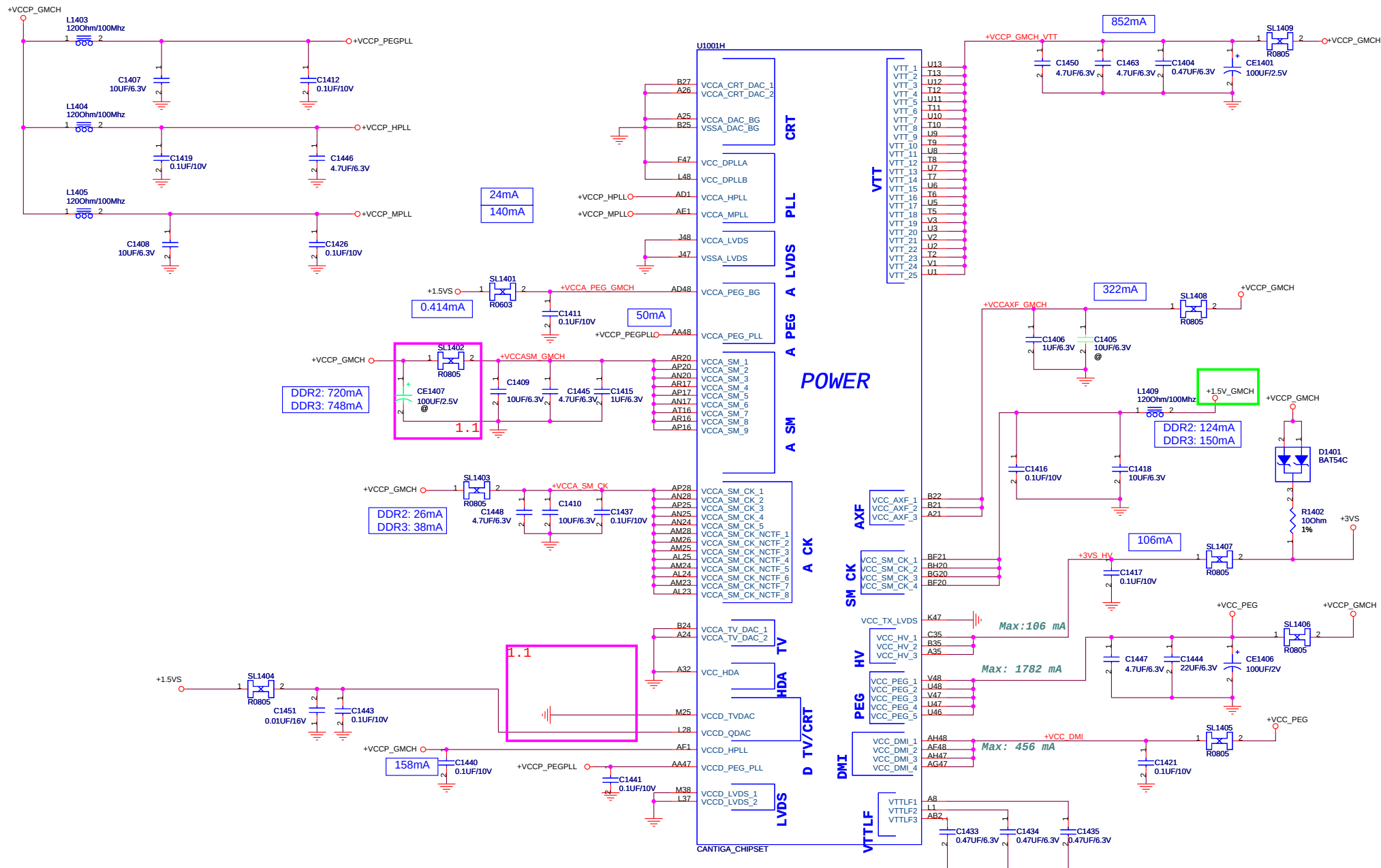
<8> M_B_DQ[0:63]

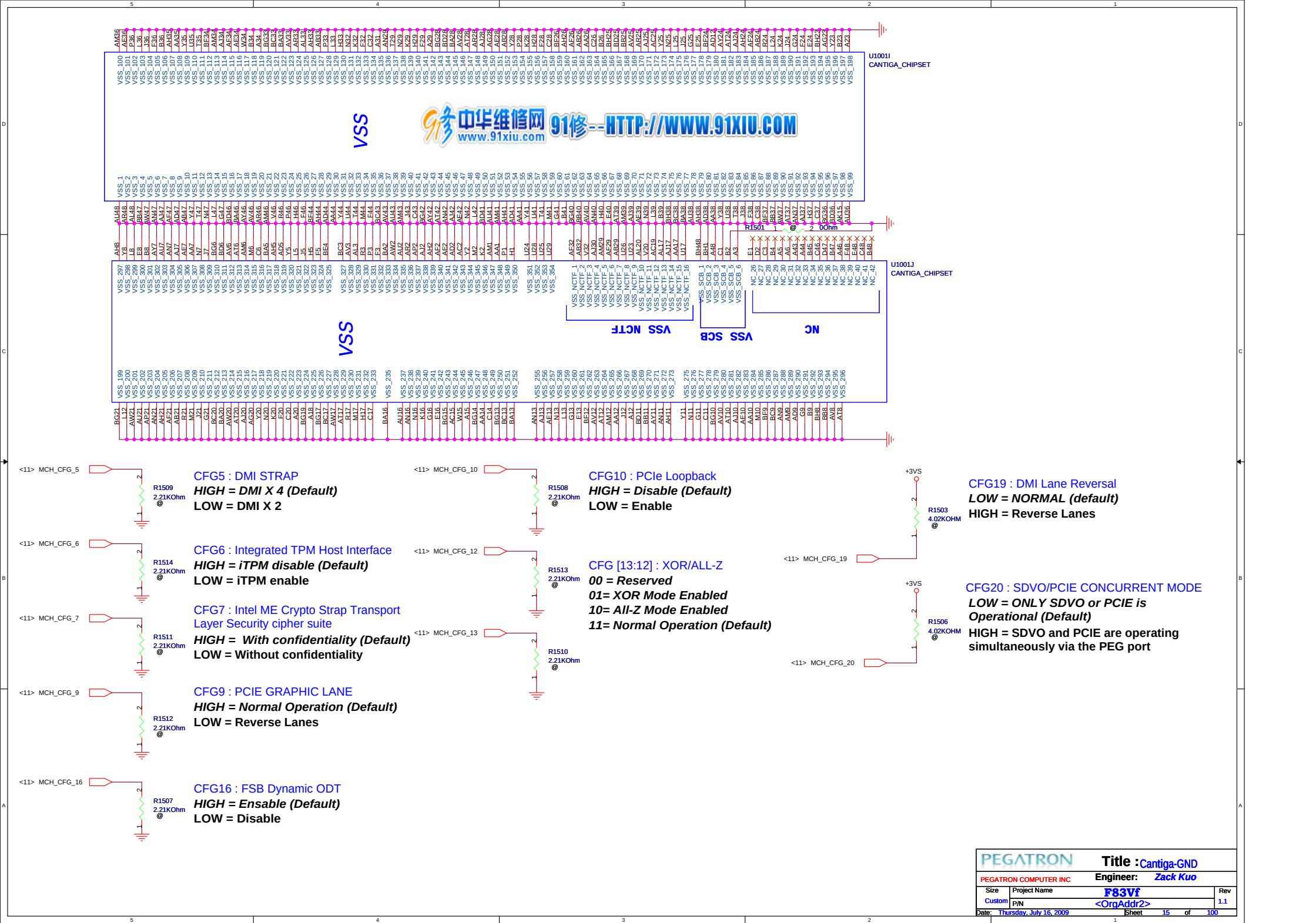


DDR SYSTEM MEMORY B

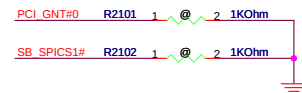
CANTIGA_CHIPSET







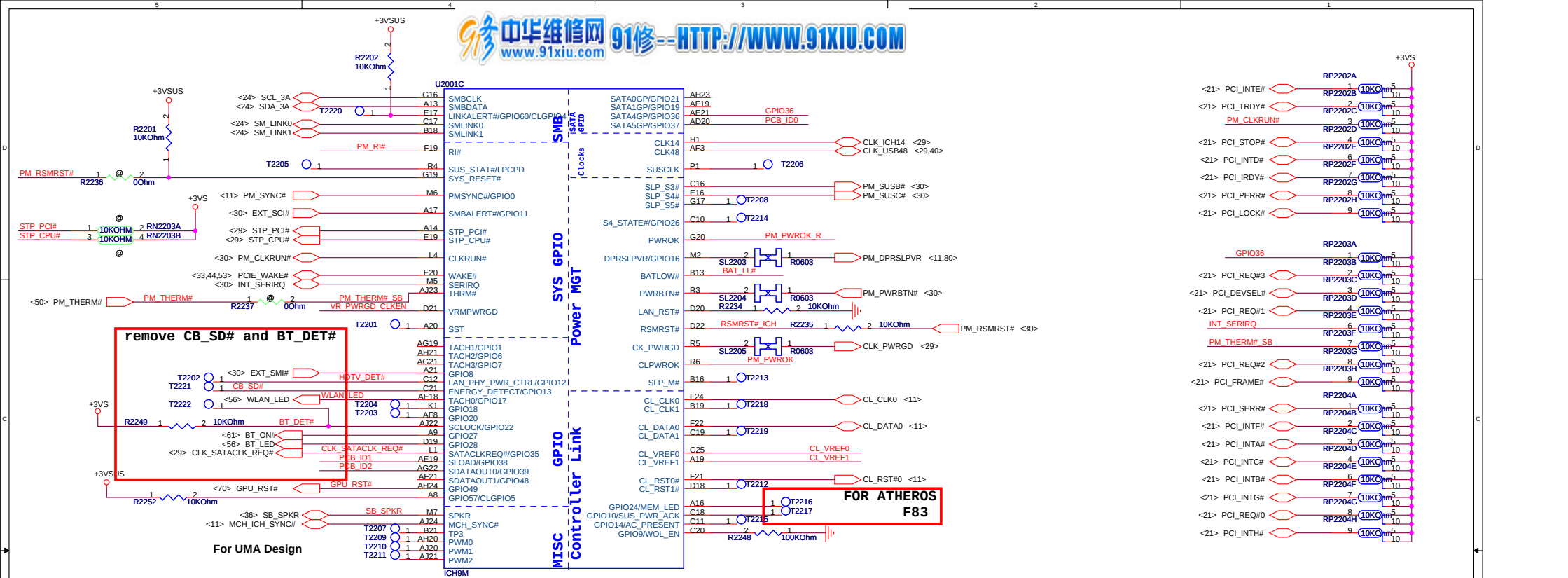
PEGATRON		Title :	
PEGATRON COMPUTER INC		Engineer: Zack Kuo	
Size	Project Name	F83Vf	Rev
Custom	P/N	<OrgAddr2>	1.1
Date: Thursday, July 16, 2009		Sheet 16 of 100	



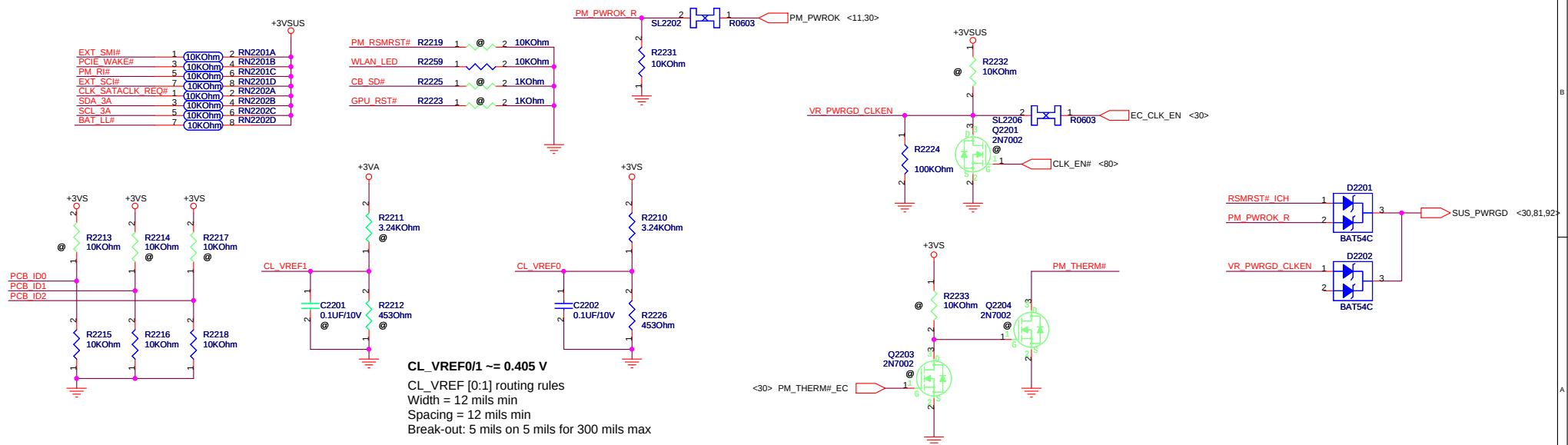
		GNT#0	CS#
LPC	11	1	1
PCI	10	1	0
SPI	01	0	1

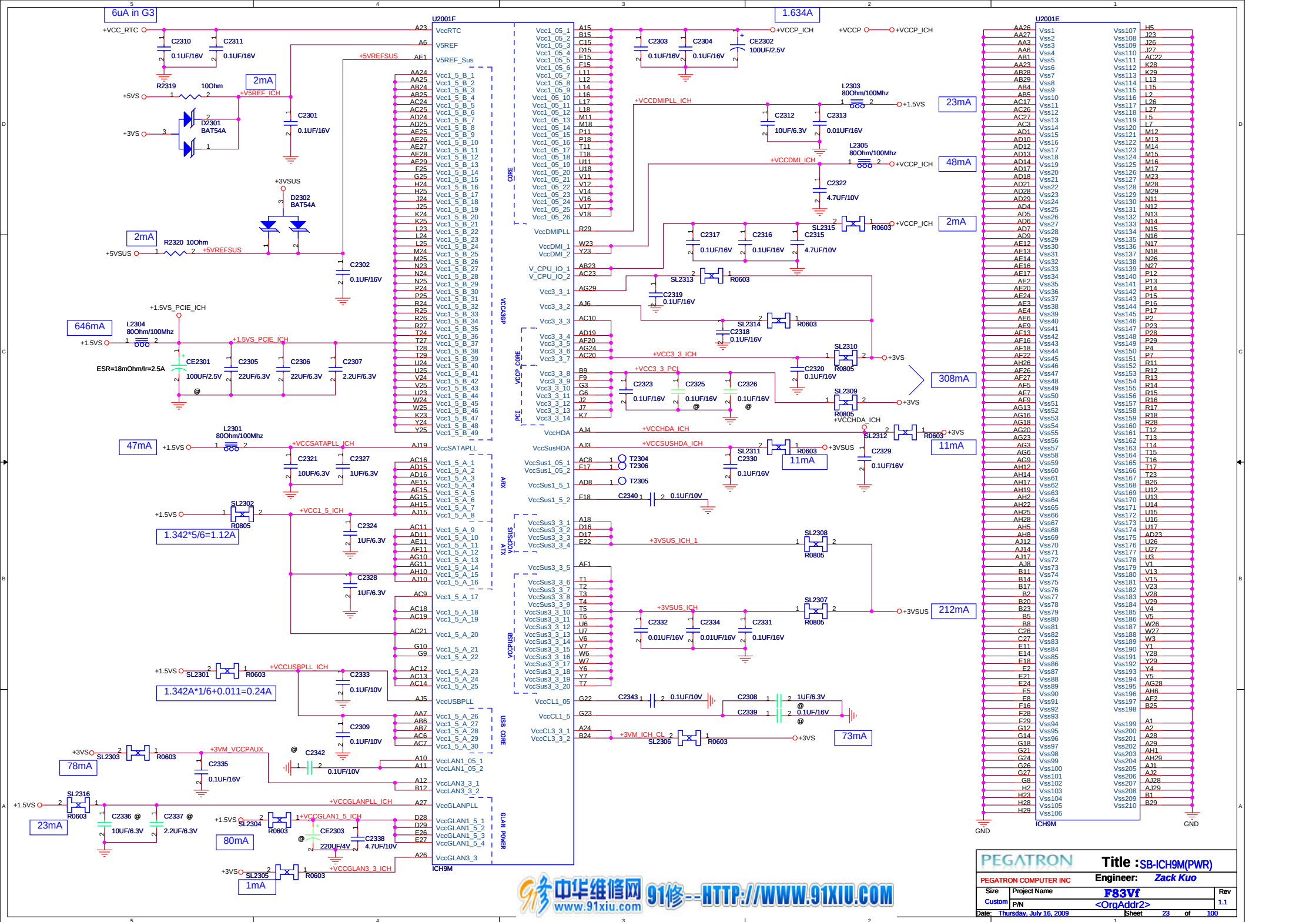
(default)

USB0	External Port
USB1	External Port
USB2	WLAN
USB3	External Port
USB4	CMOS Camera
USB5	BT
USB6	NEWCARD
USB7	CardReader
USB8	FREE
USB9	FREE
USB10	FREE
USB11	FingerPrinter

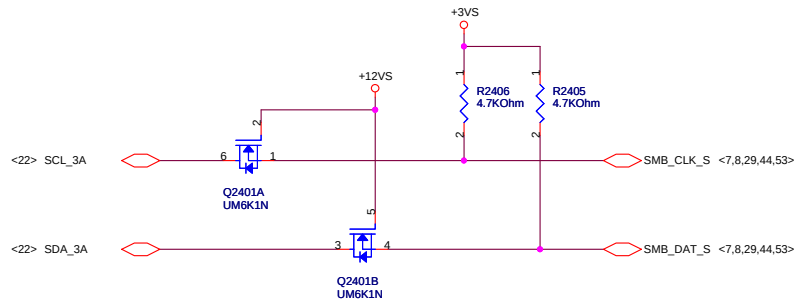


Mount/unmount as same R2236

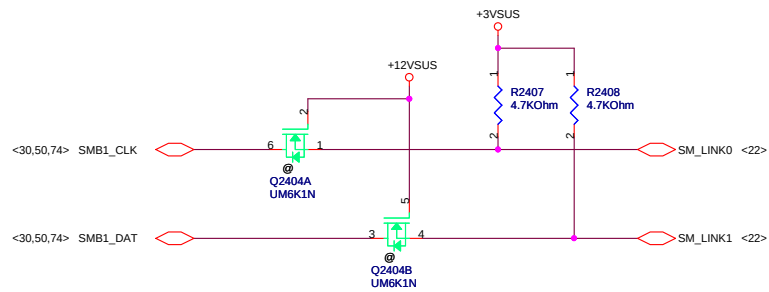


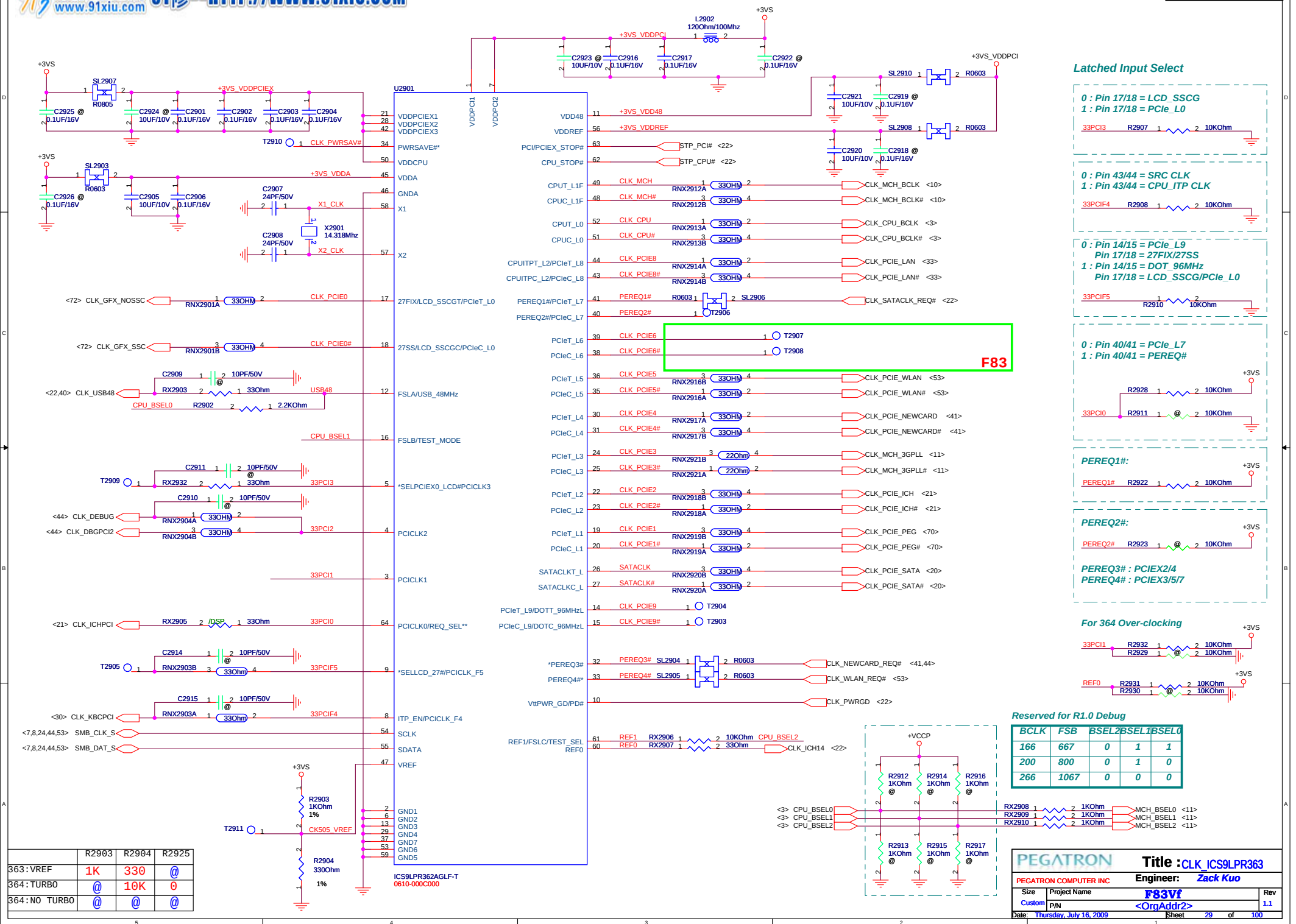


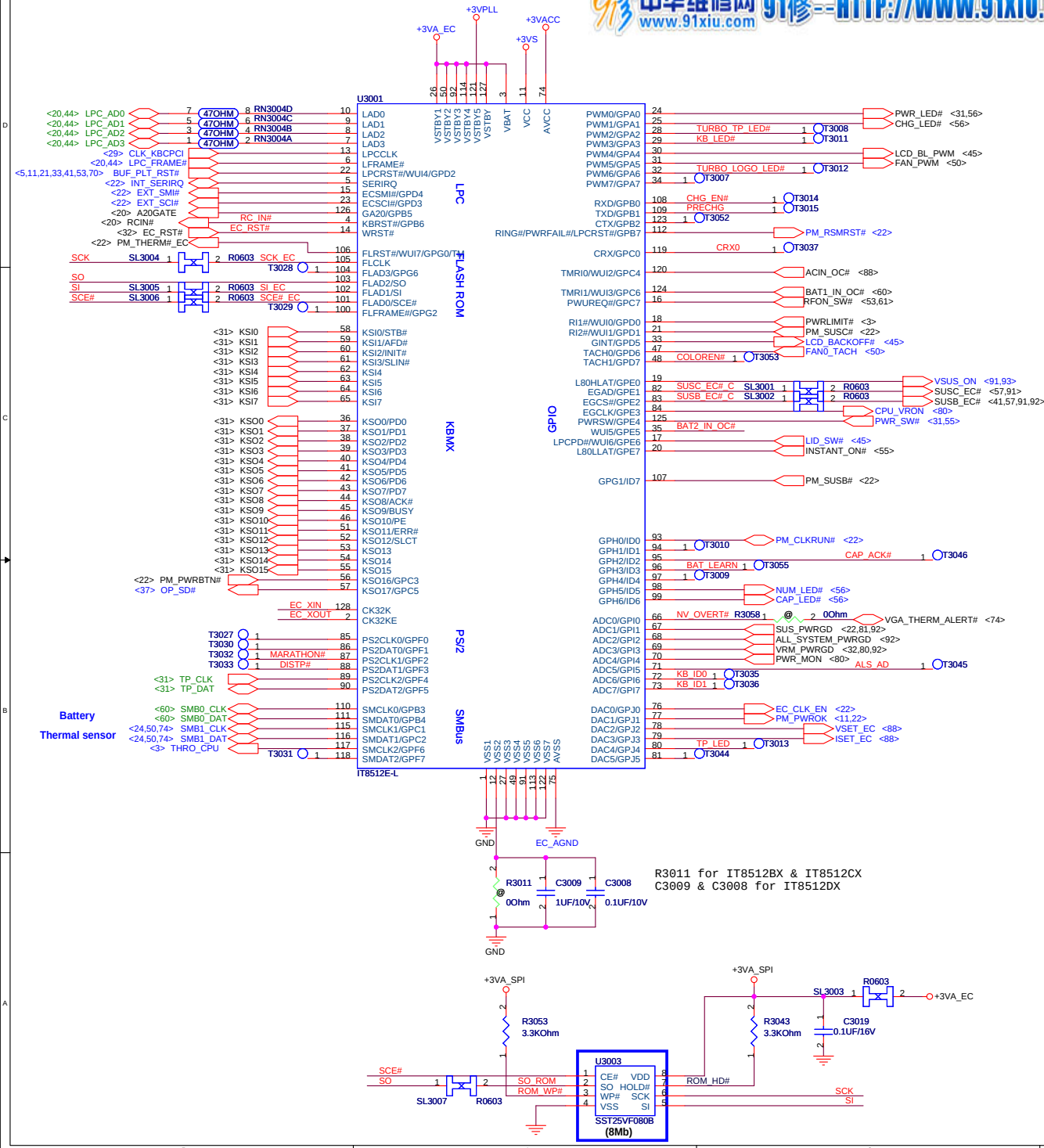
ICH9-M



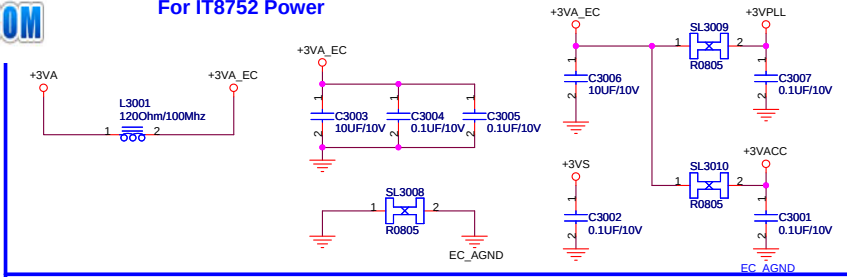
中华维修网 91修--[HTTP://WWW.91XIU.COM](http://www.91xiu.com)



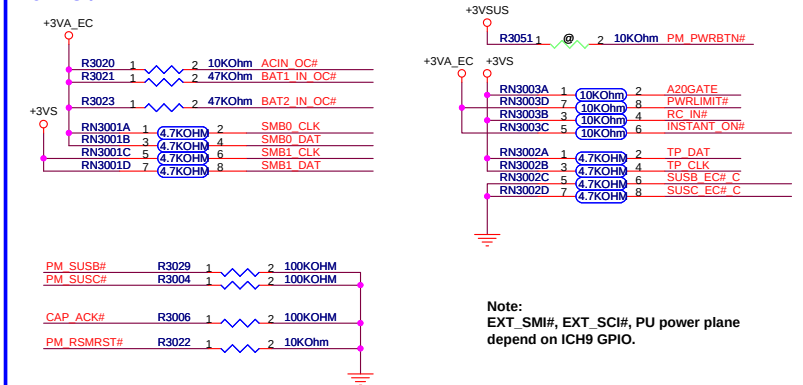




For IT8752 Power

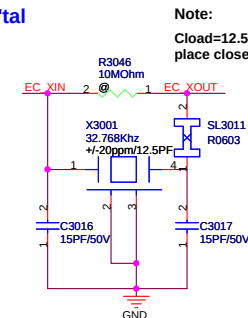


For PU / PD



Note:
EXT_SMI#, EXT_SCI#, PU power plane
depend on ICH9 GPIO.

For X'tal



Note:
Load=12.5PF
place close to EC

For EC Hardware Strap

I/O Base Address

Note: It can be programmable by EC firmware

Share Memory

Note: It can be programmable by EC firmware.

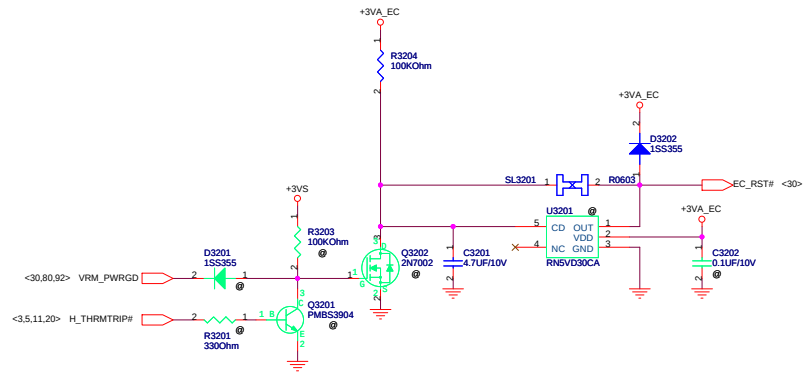
PP Enable

Note: Default Int. Pull-Low

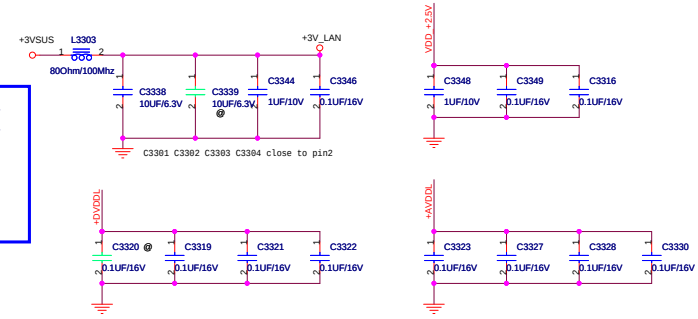
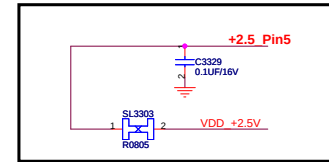
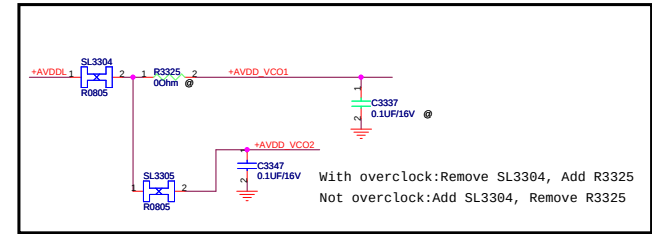
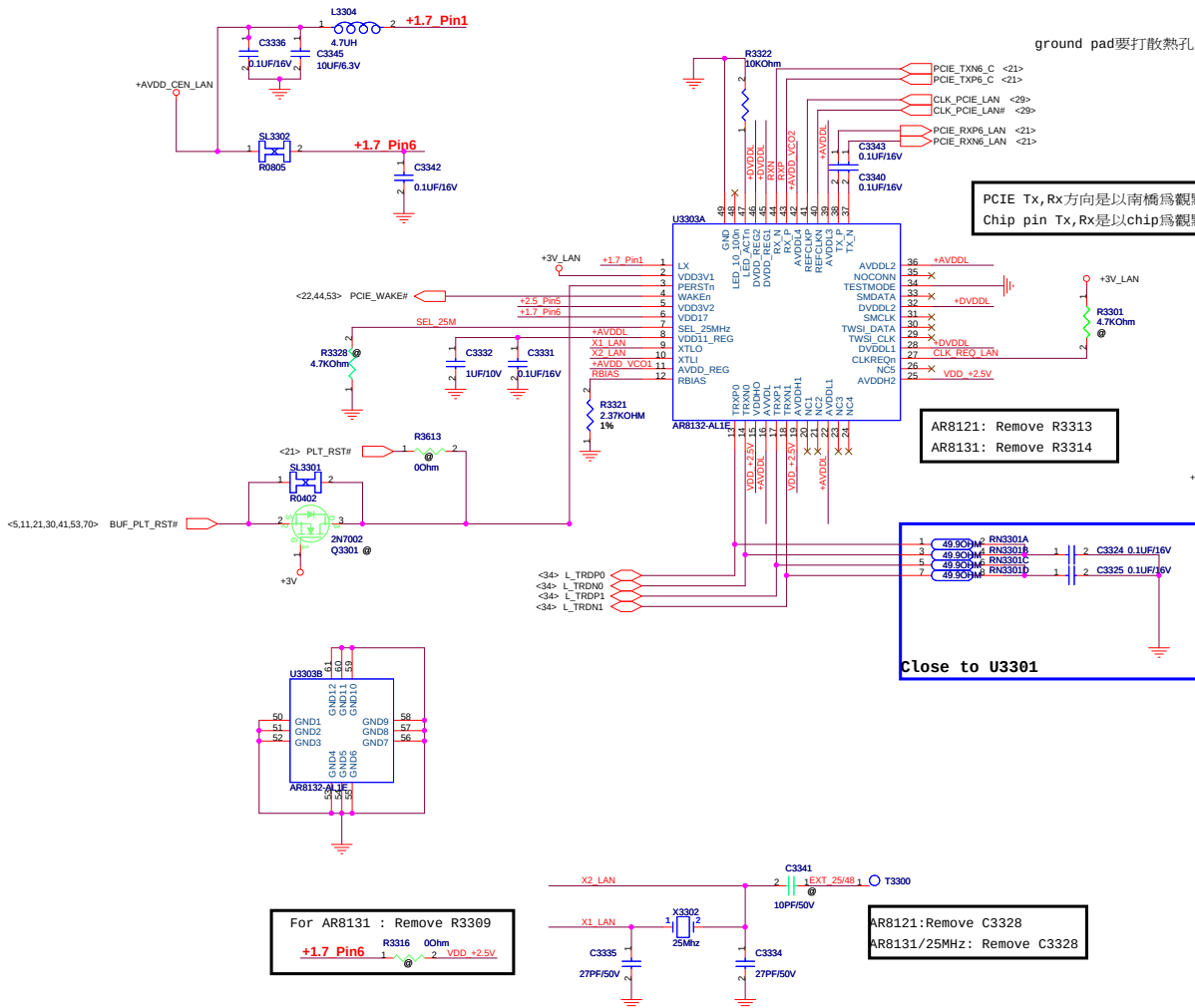
For iAMT pin name

```
AC_PRESENT
PM_S4_STATE#
S4_STATE_ON
PM_SLP_M#
SLP_M_ON
EC_WLAN_PWR
MP_PWRGD
AC_PRESENT
LAN_WOL_EN
+3VM_PG
+1.5VM_+3VMCLK_
SUSPWR_ACK
```

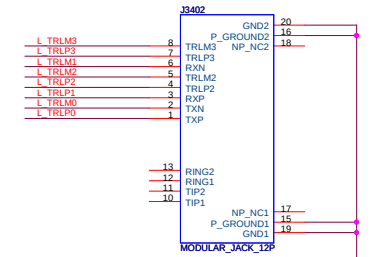
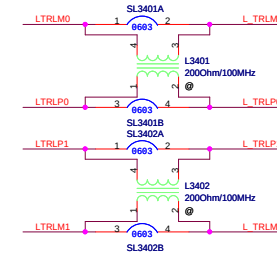
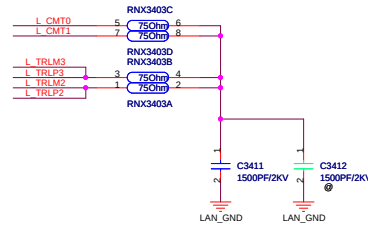
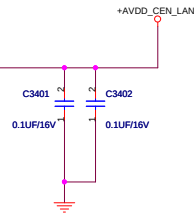
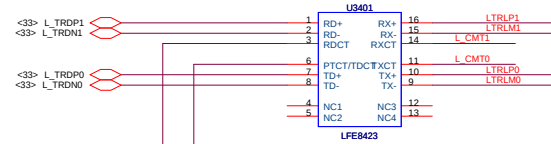
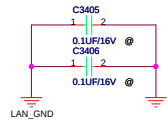
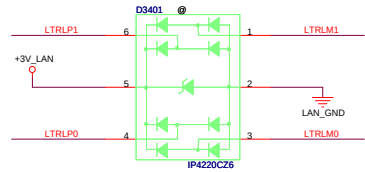
		Title : <u>EC_IT8512</u>	
PEGATRON COMPUTER INC		Engineer: <u>Zack Kuo</u>	
Size <u>Custom</u>	Project Name <u>F83vf</u> <u><OrgAddr></u>		Rev <u>1.1</u>
Date: <u>Thursday, July 16, 2009</u>		Sheet <u>30</u> of <u>100</u>	

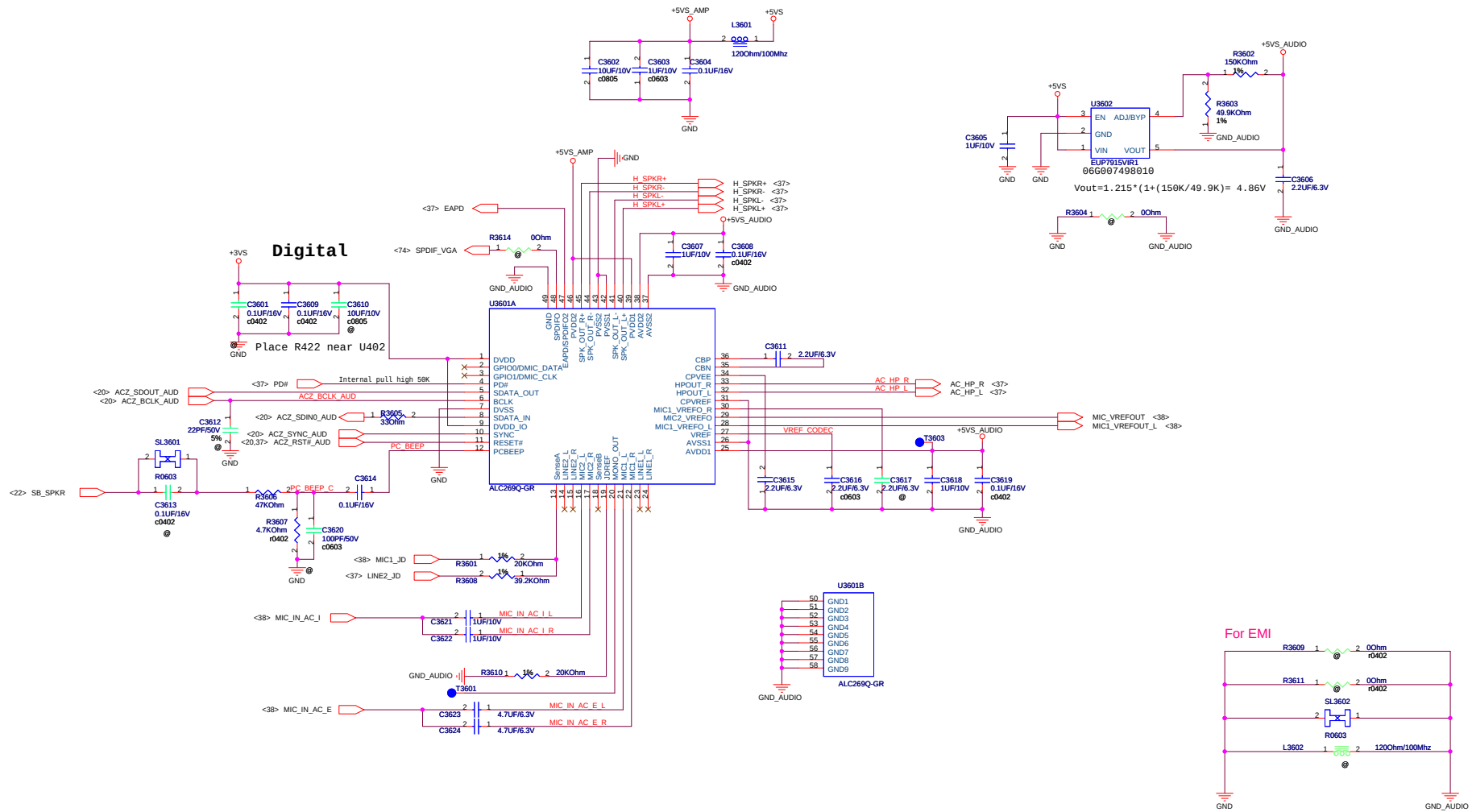


+3VSUS <3,7,8,11,14,15,22,23,24,29,30,32,36,40,41,44,45,46,48,50,51,53,56,57,70,72,74,80,91,92>
+3VSUS <20,21,22,23,24,30,37,45,53,56,91,92>
GND <3,4,5,7,8,9,10,11,13,14,15,20,21,22,23,29,30,31,32,34,36,37,38,40,41,44,45,46,48,50,51,52,53,55,56,57,60,61,63,68,70,72,73,74,75,76,77,80,81,82,83,84,85,88,91,92>
LAN_GND <34>
LAN_GND

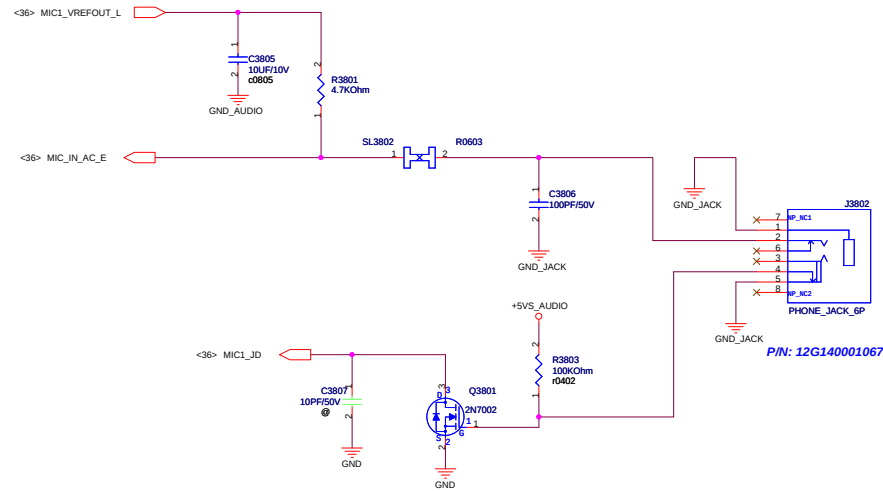
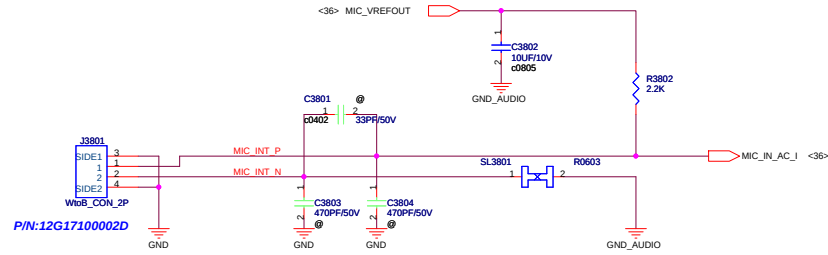


LAN / MODEM PORT

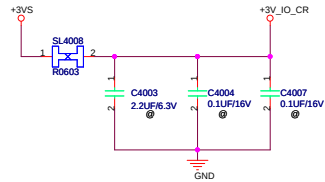




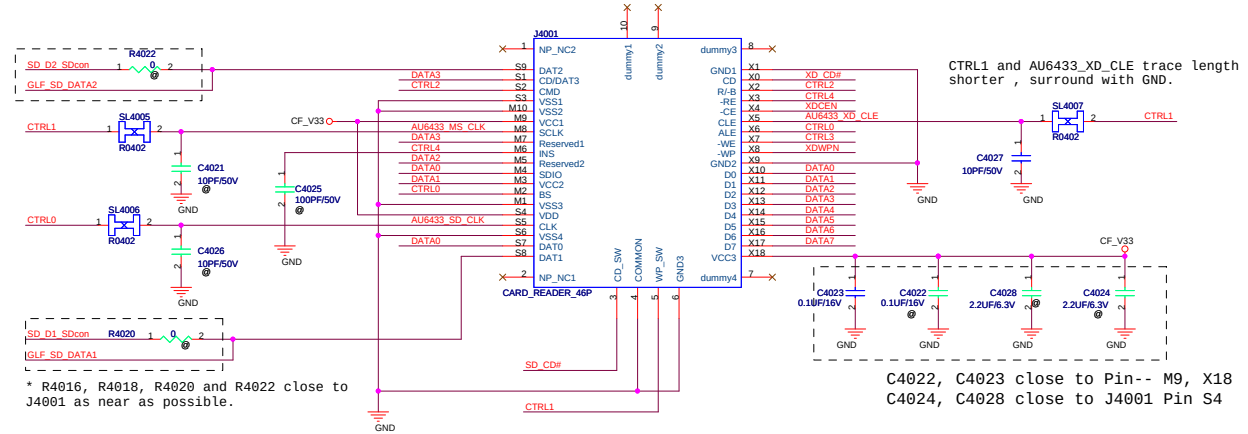
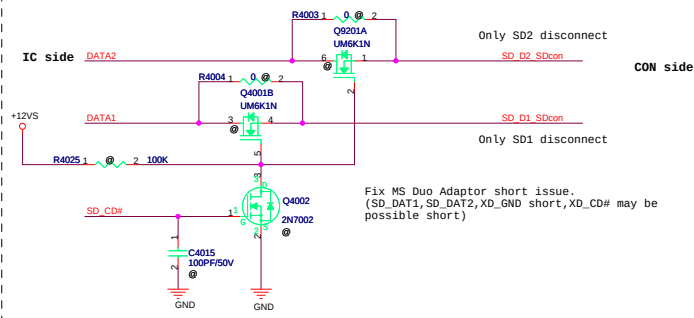
Internal MIC Pre-Amplifier



CARD READER CONNECTOR

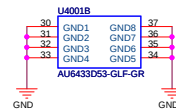


For AU6433-GEF

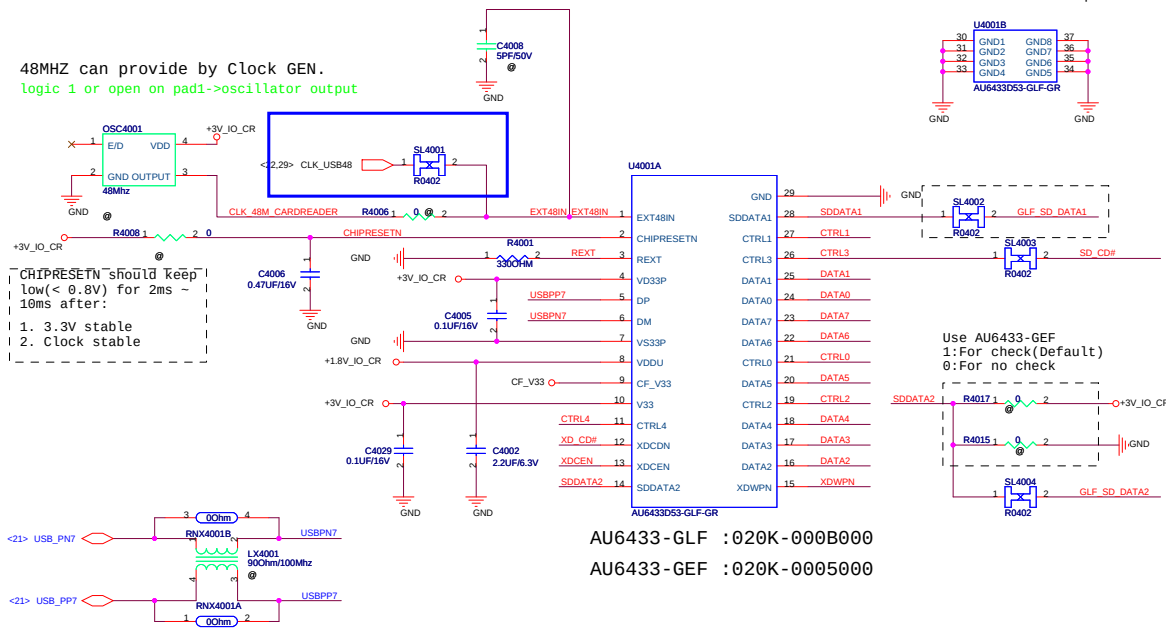


C4022, C4023 close to Pin-- M9, X18
C4024, C4028 close to J4001 Pin S4

0304 for card reader chip via



48MHZ can provide by Clock GEN.
logic 1 or open on pad1->oscillator output



```
CTRL0->SDCLK/XDALE/MSBS
CTRL1->SDWP/XDCLK/MSCLK
CTRL2->SDCMD/XDRBN
CTRL3->SDCDN/XDWRN
CTRL4->XDRDN/MSINS

XDCDN->XDCDN
XDCEN->XDCEN
SDDATA2->SDDATA2/xD CIS check is disable
XDWPN->XDWPN

ADDATA1->SDDATA1

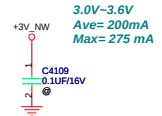
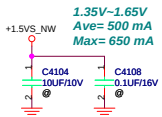
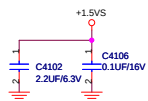
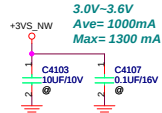
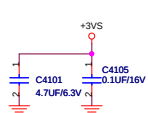
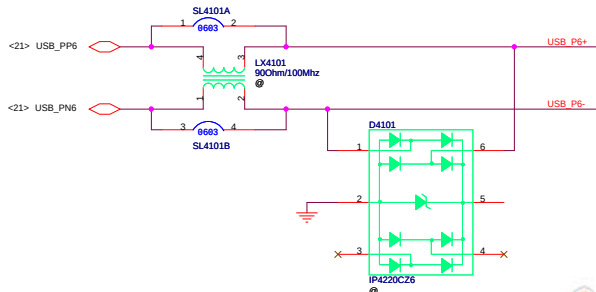
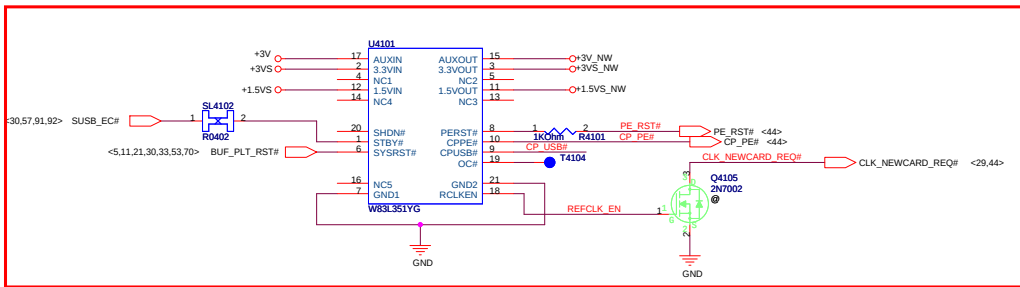
DATA0->SDDATA0/XDDATA0/MSDATA0
DATA1->SDDATA1/MSDATA1
DATA2->XDDATA2/MSDATA2
DATA3->SDDATA3/XDDATA3/MSDATA3
DATA4->SDDATA4/XDDATA4/MSDATA4
DATA5->SDDATA5/XDDATA5/MSDATA5
DATA6->SDDATA6/XDDATA6/MSDATA6
DATA7->SDDATA7/XDDATA7/MSDATA7
```

AU6433 GEF /GLF colay.

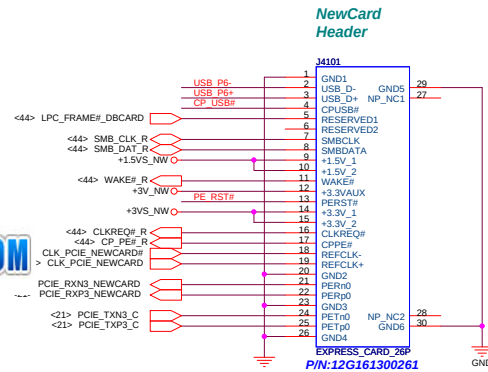
Option 1: AU6433-GEF
If use GEF package need:

```
unmount-> R4016,R4018
mount->R4017,R4020,R4022,R4025,Q4001,Q4002.
```

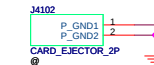
```
Option 2:AU6433-GLF
If use GLF package need:
unmount->
R4017,R4020,R4022,R4025,Q4001,Q4002.
mount->R4016,R4018
```

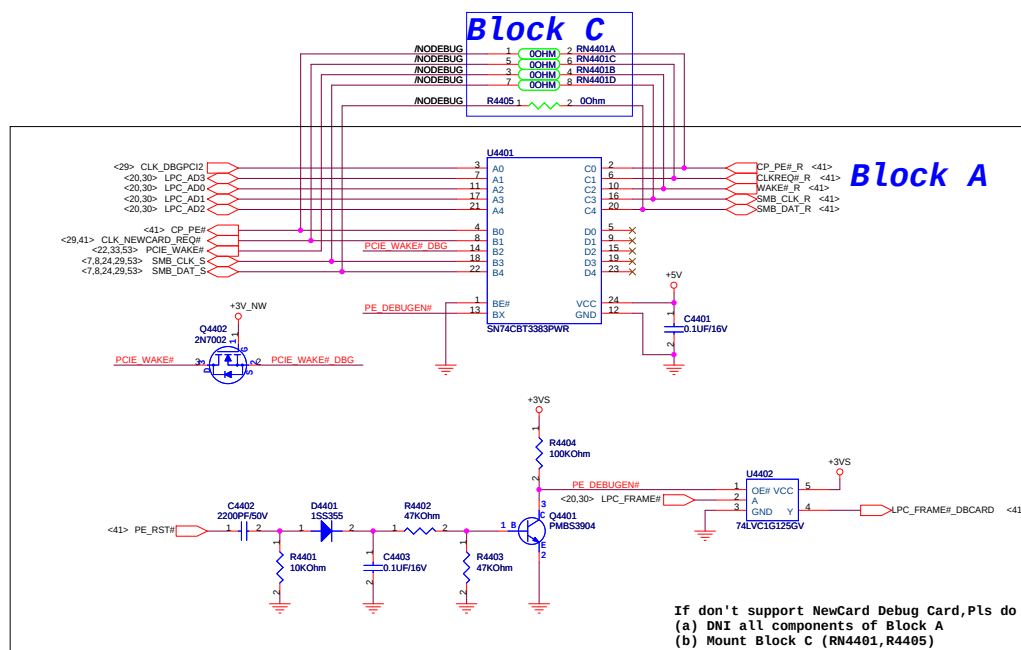
9修 中华维修网 91修 -- [HTTP://WWW.91XIU.COM](http://www.91xiu.com)



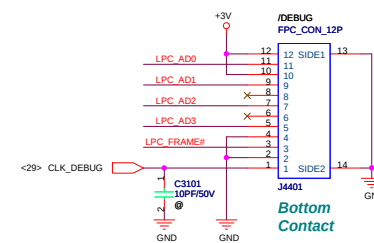
NewCard Ejecutor



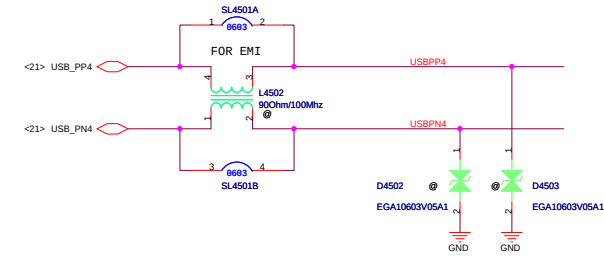
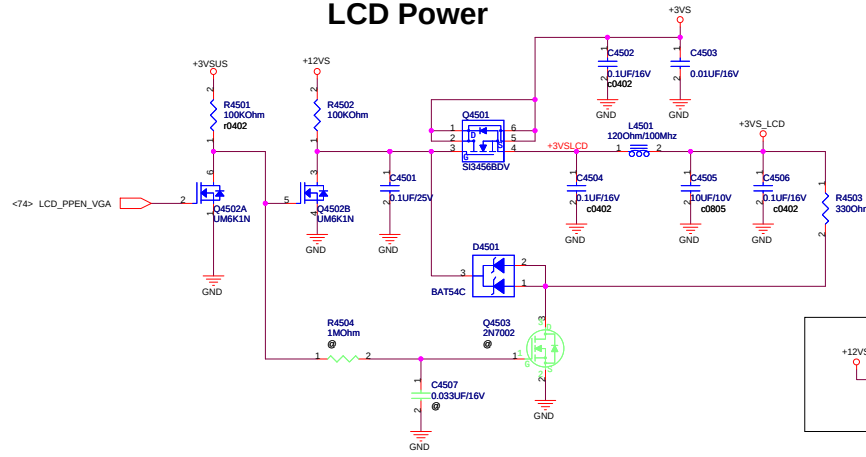
For NewCard Debug Card



LPC Debug Port

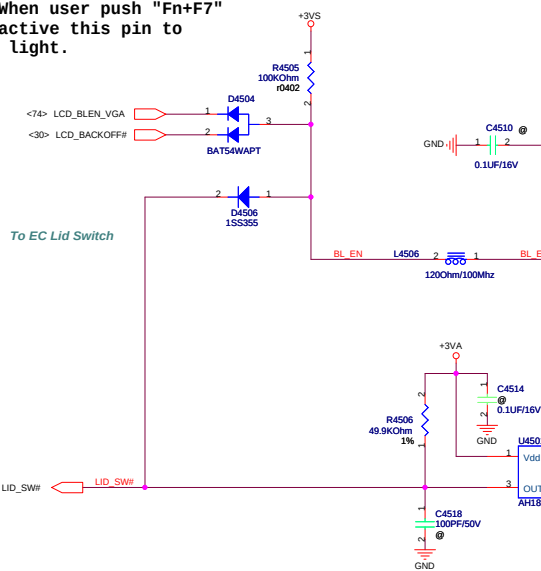


LCD Power

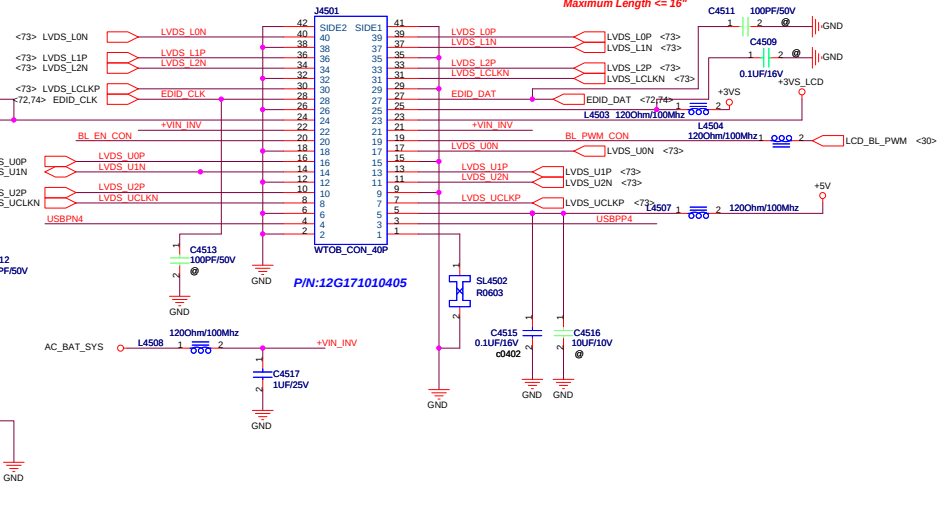


BIOS

LCD_BACKOFF#: When user push "Fn+F7" button, BIOS active this pin to turn off back light.

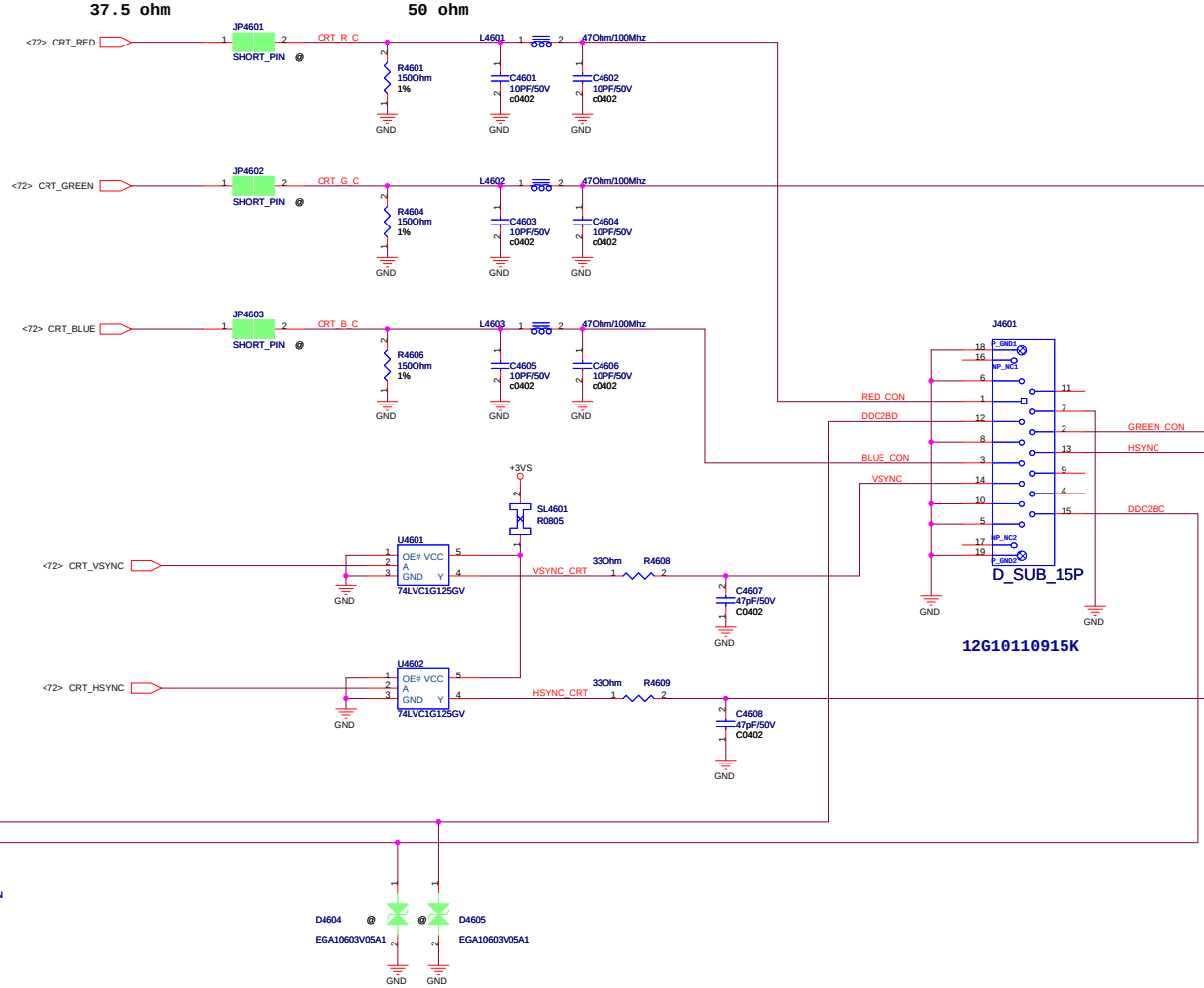
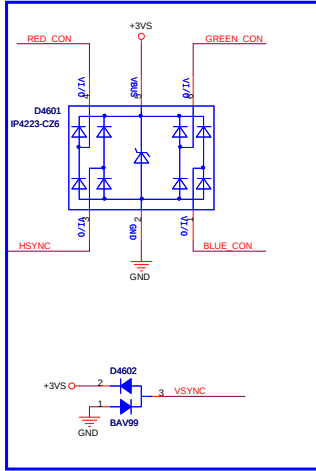


LCD LVDS/Inverter/CCD conn.



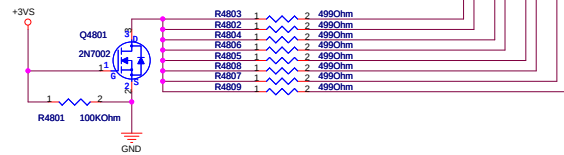
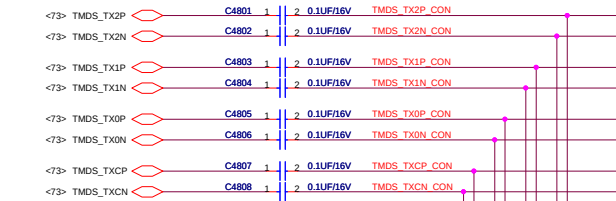
Cable Requirement:
Impedance: 100 ohm +/- 10%
Length Mismatch <= 10 mils
Twisted Pair (Not Ribbon)
Maximum Length <= 16"

PLACE ESD Diodes near connector

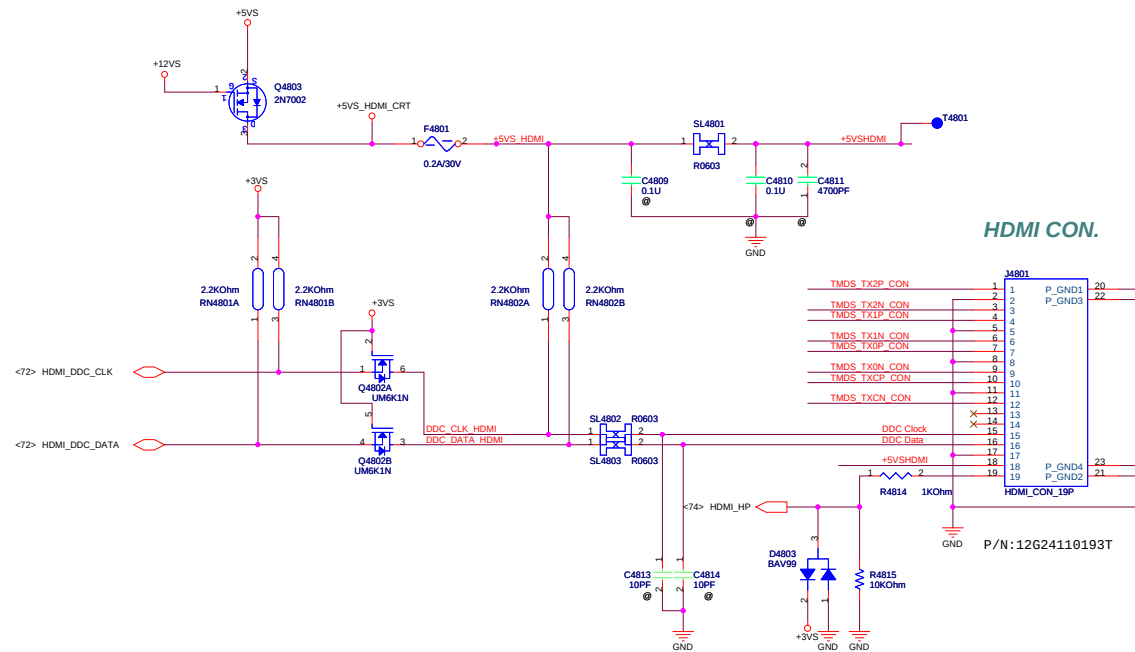


HDMI

near the HDMI connector

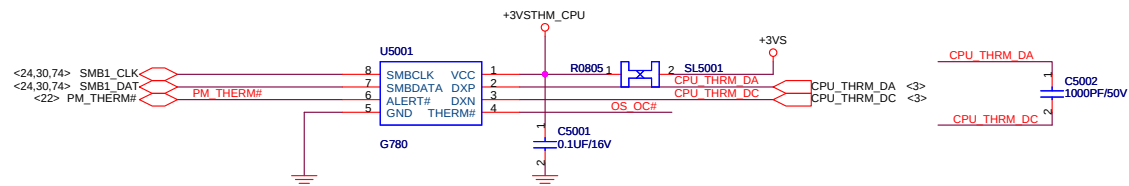


Reference should be +5VS, but AT answer that +3VS is fine. As long as it can turn the MOSFET on.

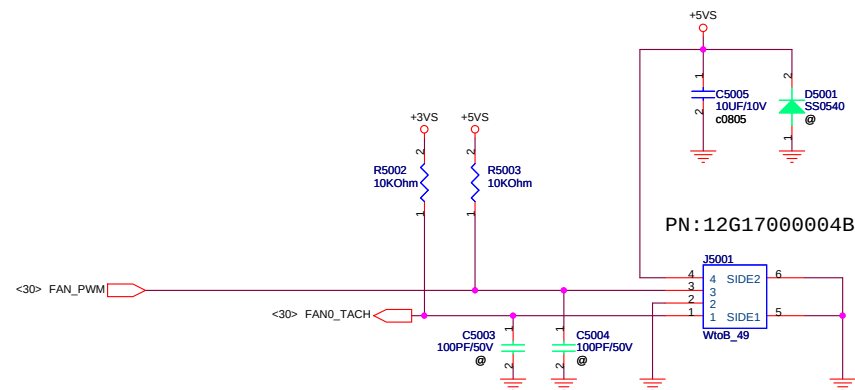


Note: 1. L1805, L1806, L1807: For EMI. (default=0 ohm)
2. DDC_CLK_HDMI, DDC_DATA_HDMI: +5V tolerant

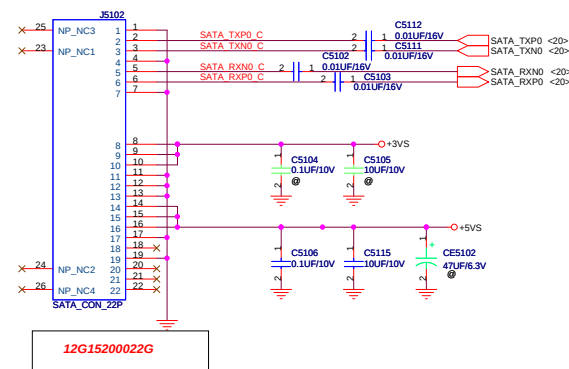
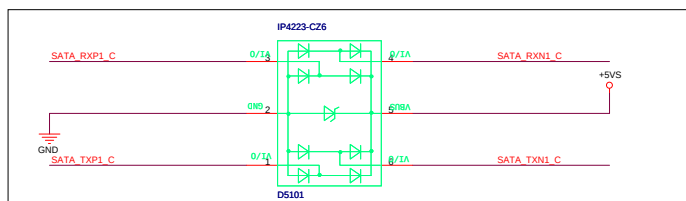
CPU Thermal Sensor



PWM Fan

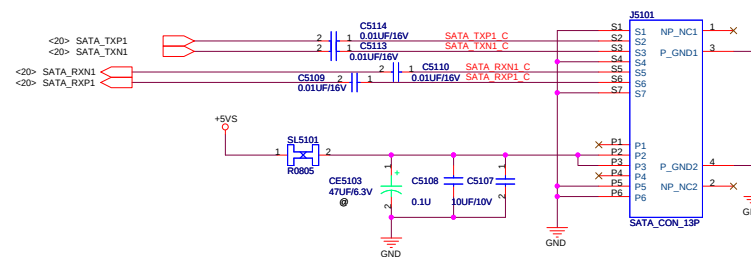


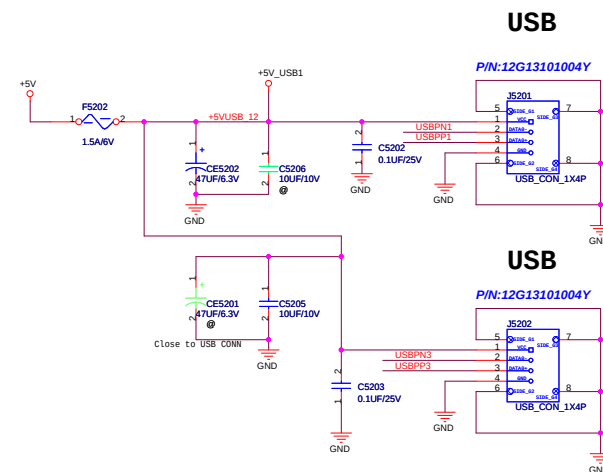
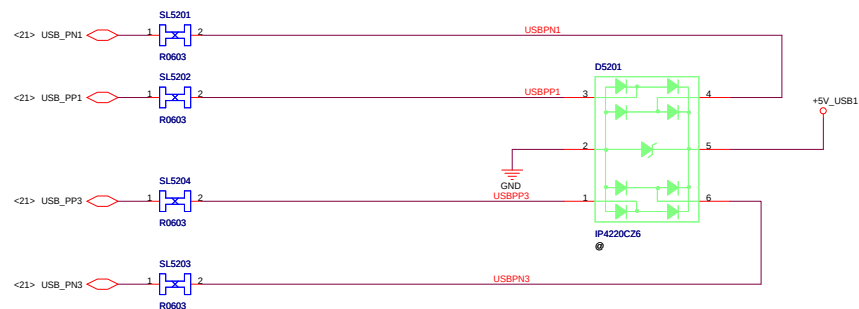
SATA HDD con.



SATA CD-ROM con.

P/N:12G15100013J

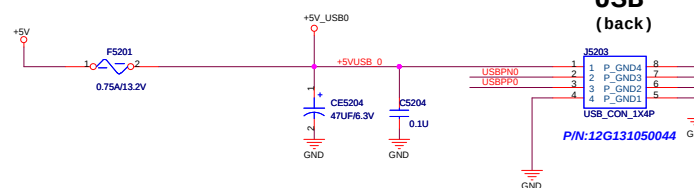
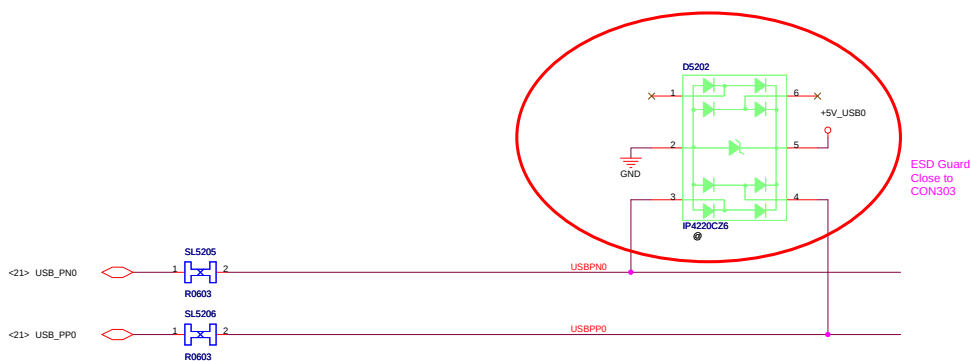


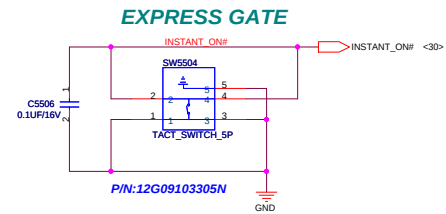
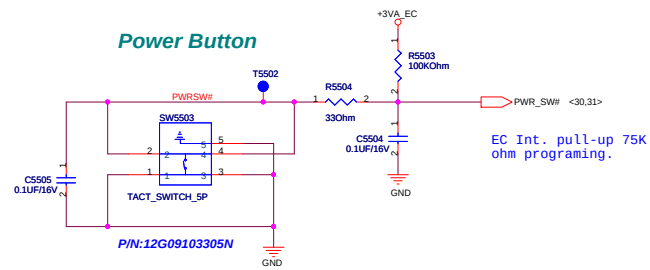


USB

USB

USB
(back)



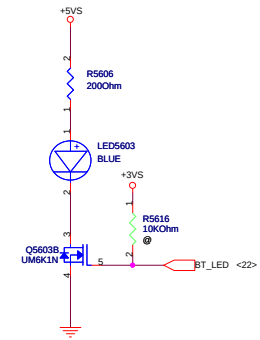
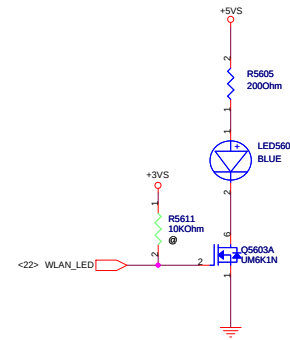
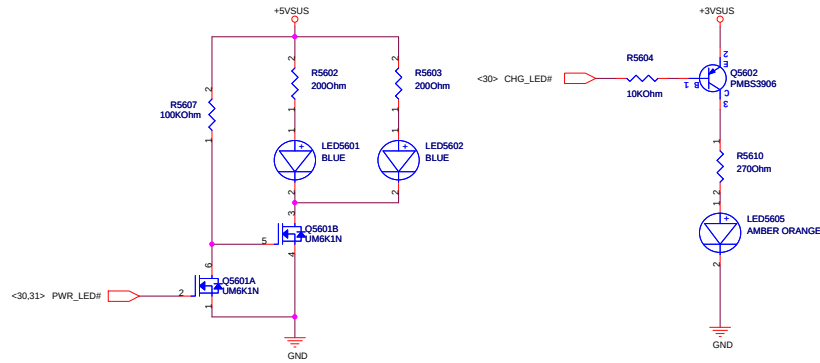


PWR LED

For BATTERY LED

WireLess LED

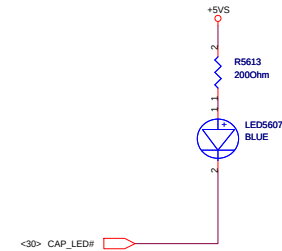
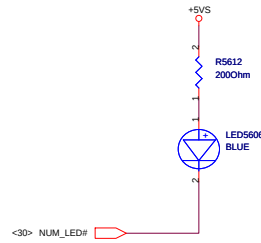
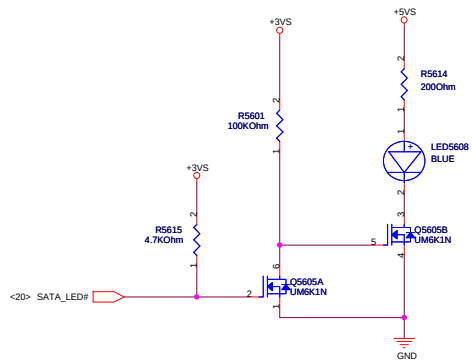
BT LED

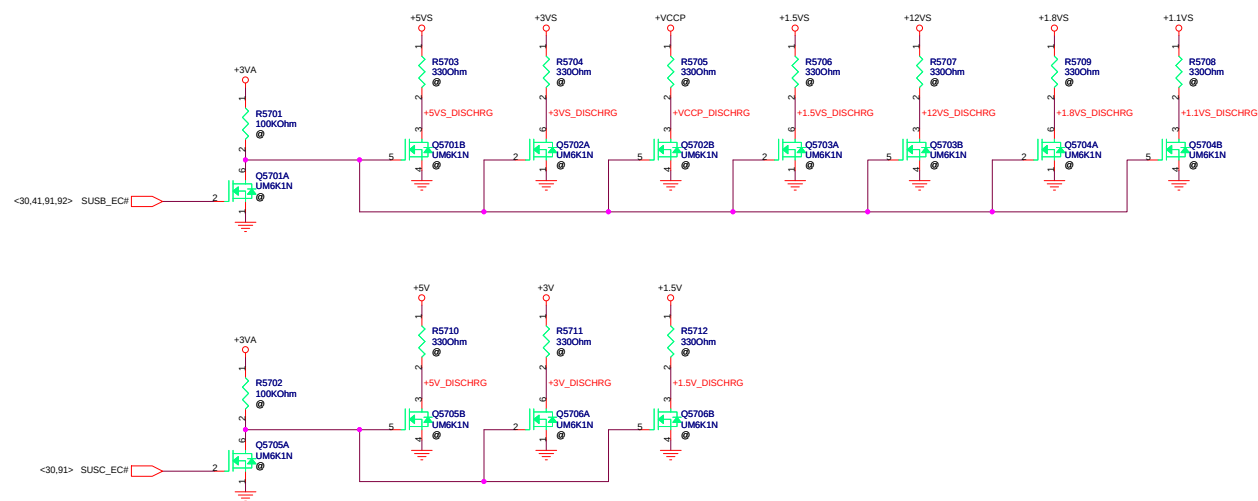


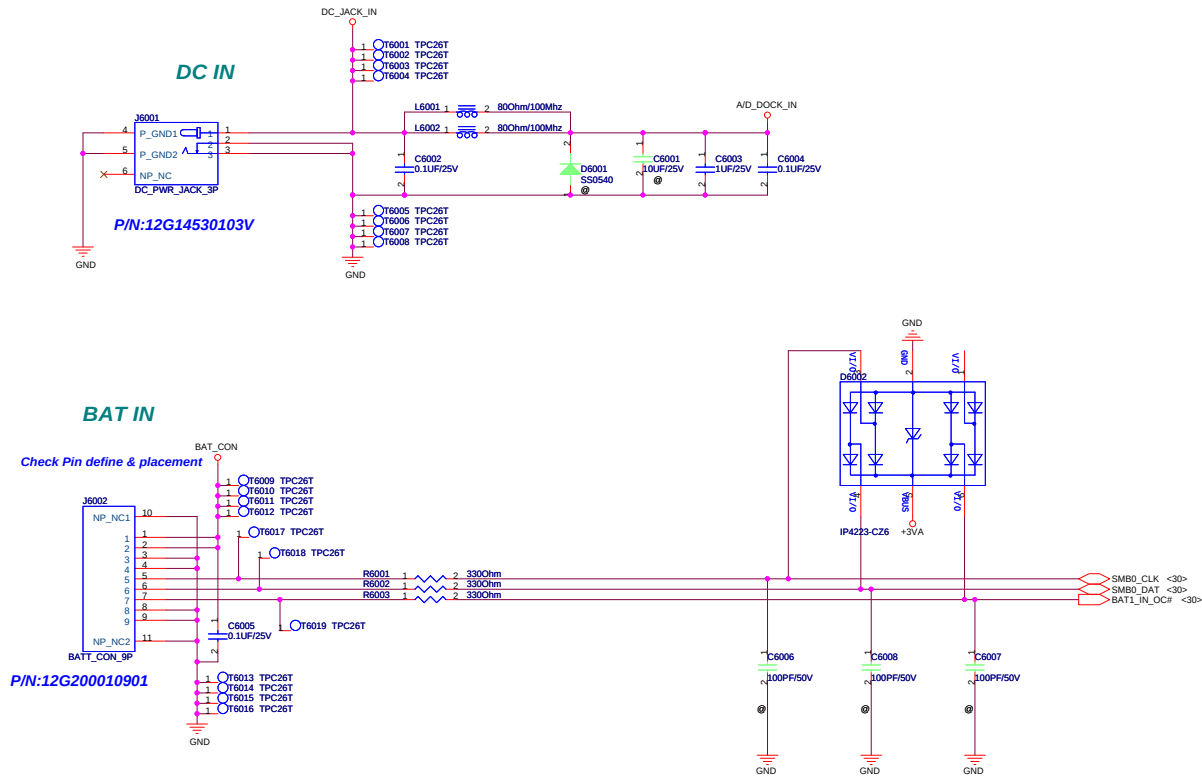
SATA LED

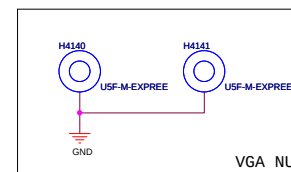
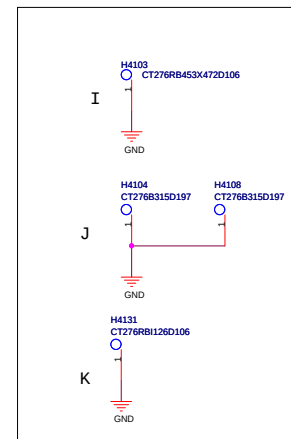
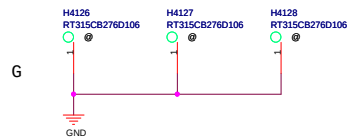
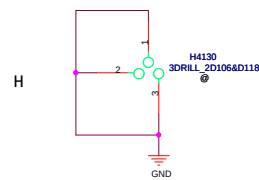
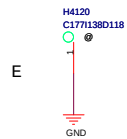
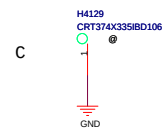
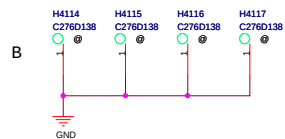
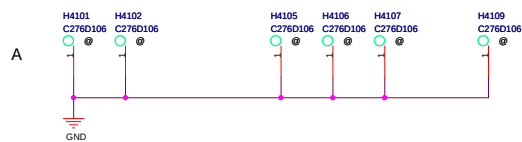
Num Lock

Cap. Lock









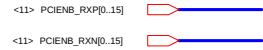
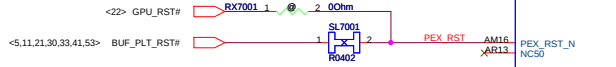
R1.1

2009/06/30

- 1. Set DDR3 VREF to 0.75V LDO output.
- 2. Change Card Reader to AU6433D53-GLF.
- 3. Change LAN to Atheros AR6132.
- 4. Change Transformer to 10/100 TAIMAG HA003
- 5. Change ClockGen to ICS9LPR363.
- 6. Remove ClockGen 3362 circuits.
- 7. Unstuff Finger Printer Connector.
- 8.

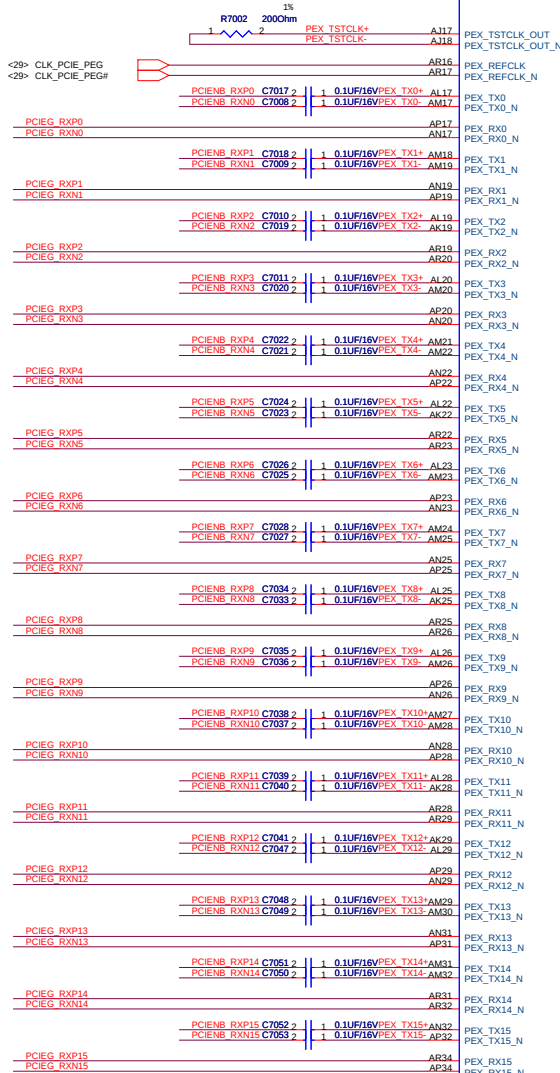


PEGATRON		Title : Connector, LED	
PEGATRON COMPUTER INC		Engineer: Zack Kuo	
Size	Project Name	F83Vf	Rev
C	P/N	<OrgAddr2>	1.1
Date: Thursday, July 16, 2009		Sheet	69 of 100

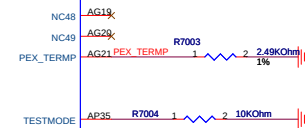
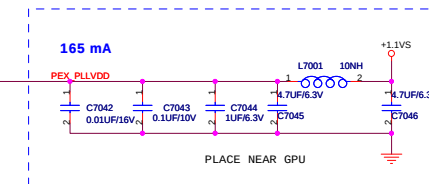
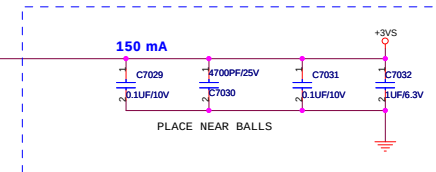
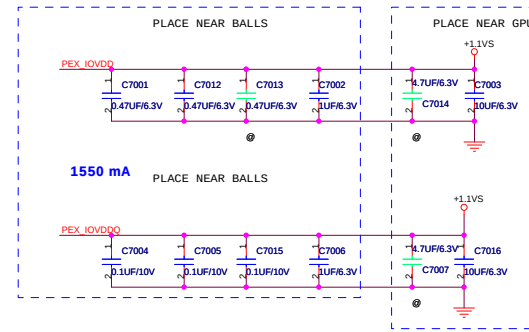
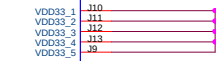
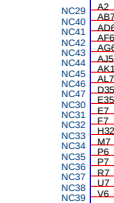
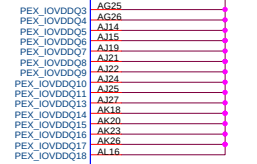
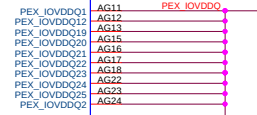
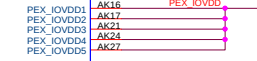


PEX=> From NB

EXP: VGA Card to NB



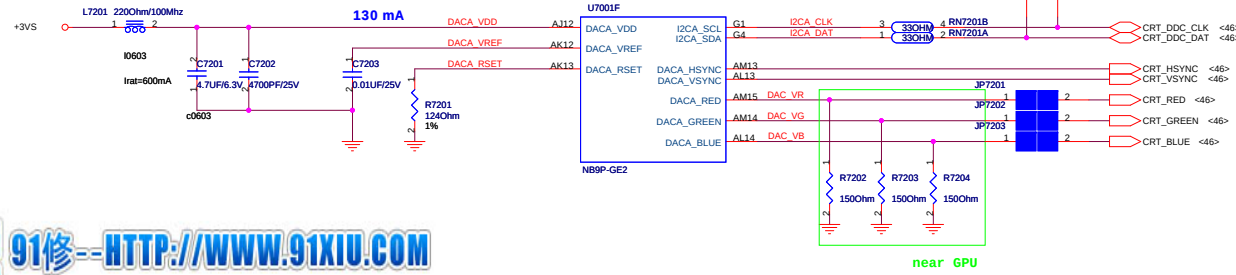
U7001A



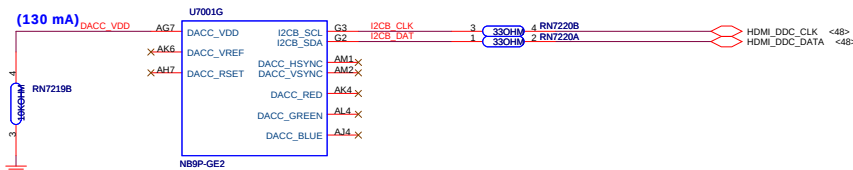
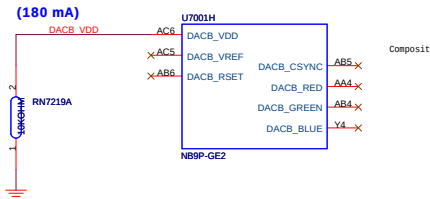
NB9P-GE2

change 4.7uF to 1uF
C5001, C5005, C5015

CRT

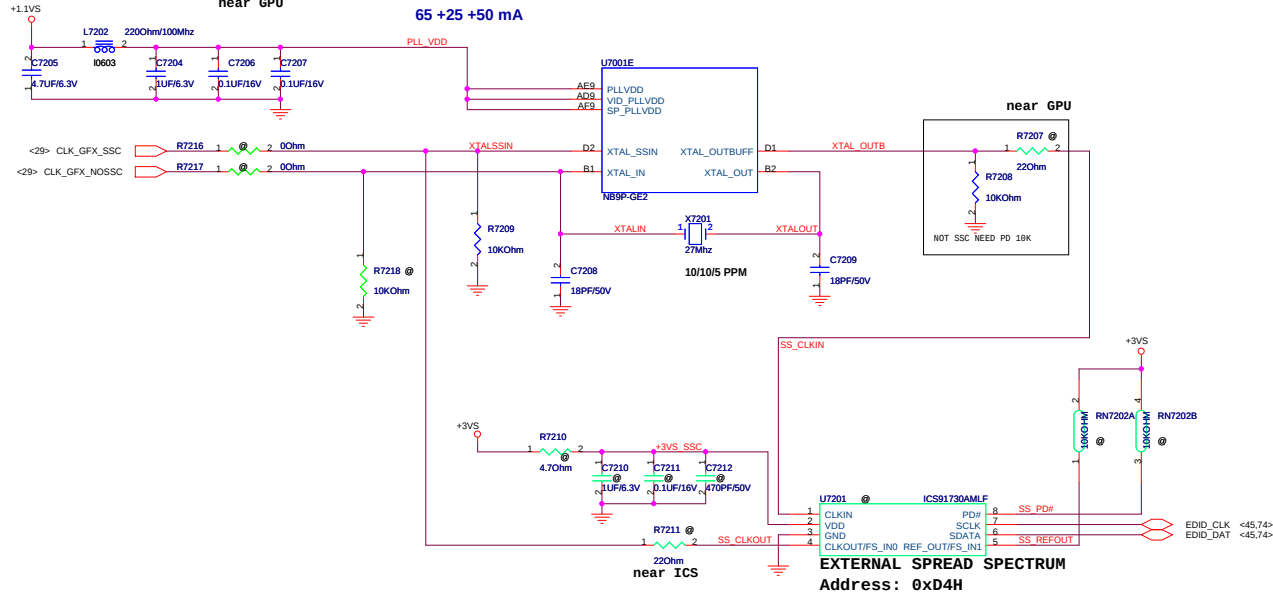


91修 中华维修网 91修 == HTTP://WWW.91XIU.COM



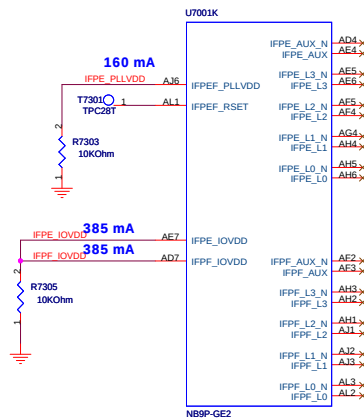
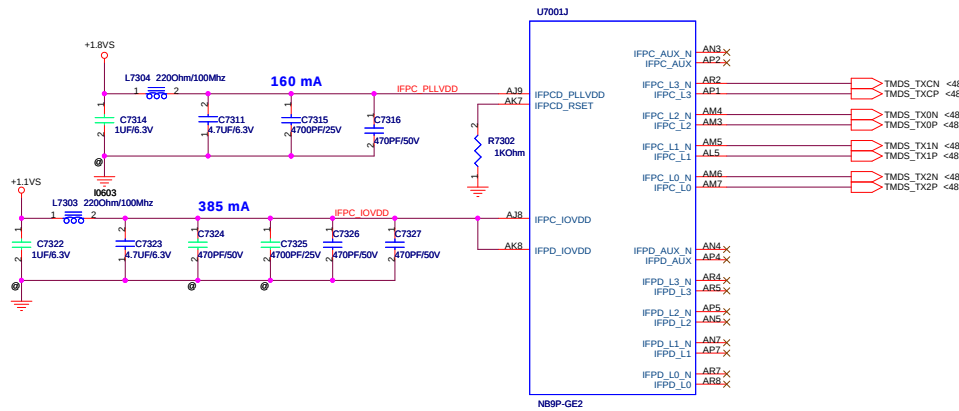
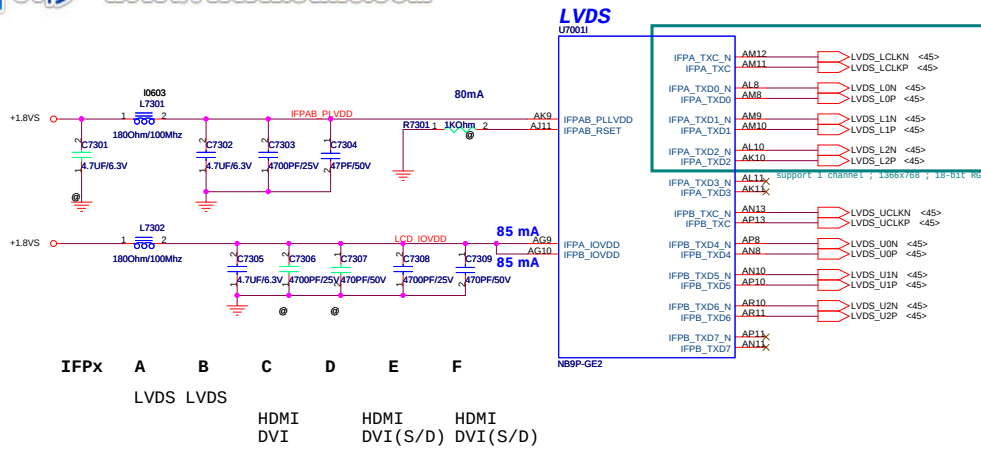
XTAL_IN, XTAL_OUT
3.3V tolerance

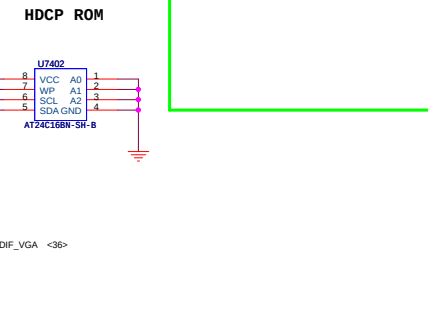
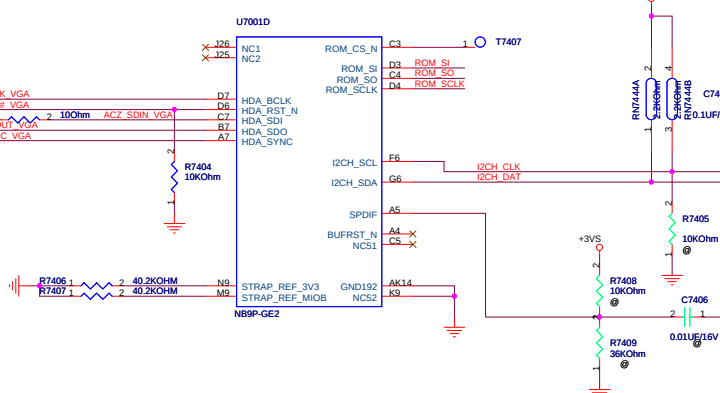
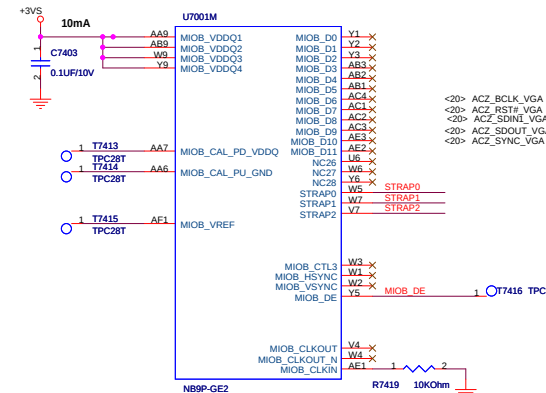
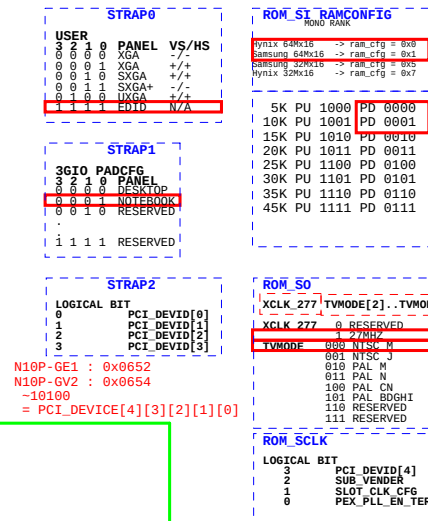
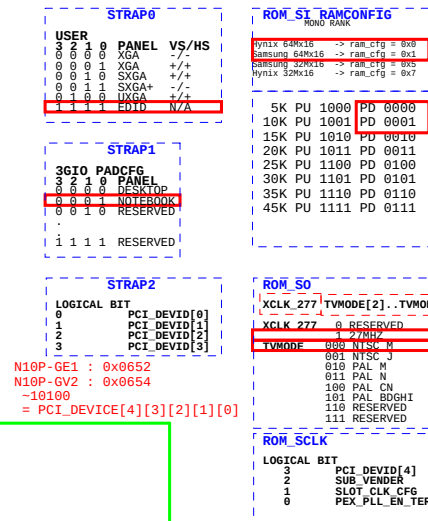
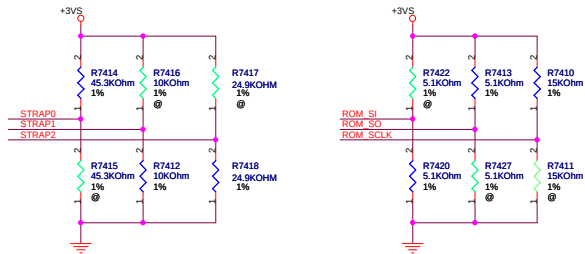
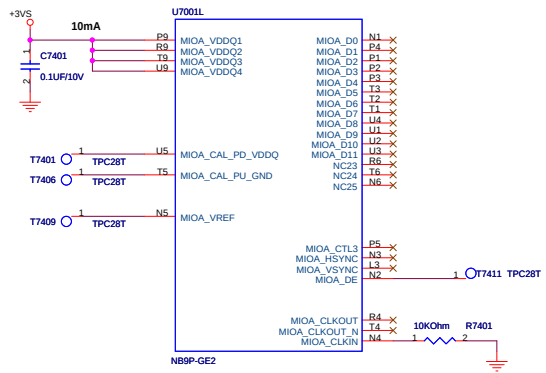
correspondent BGA balls must be
12mils and 16 mil wide



EXTERNAL SPREAD SPECTRUM
Address: 0x04H

PEGATRON		Title : Connector, LED	
PEGATRON COMPUTER INC		Engineer: <OrgAddr1>	
Size	Project Name	F83Vf	Rev
C	PIN	<OrgAddr2>	1.1
Date: Thursday, July 16, 2009		Sheet 72 of 100	

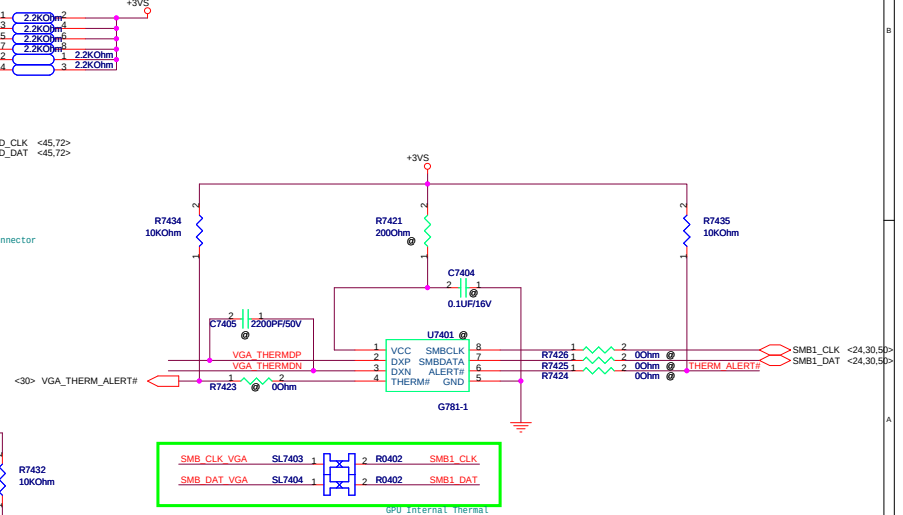
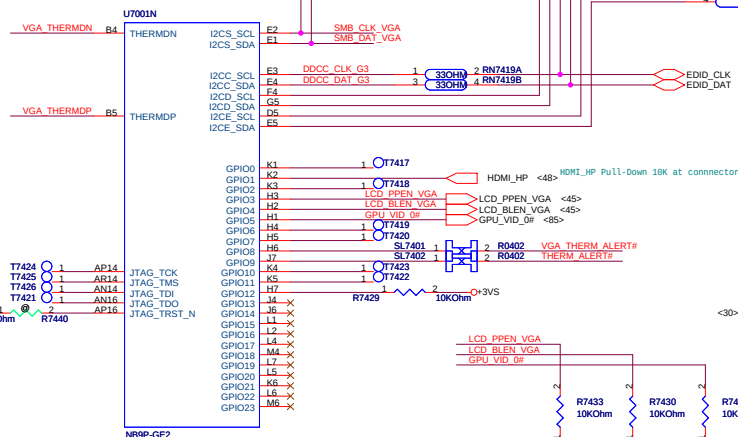


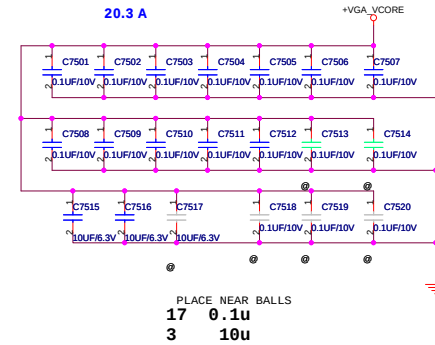
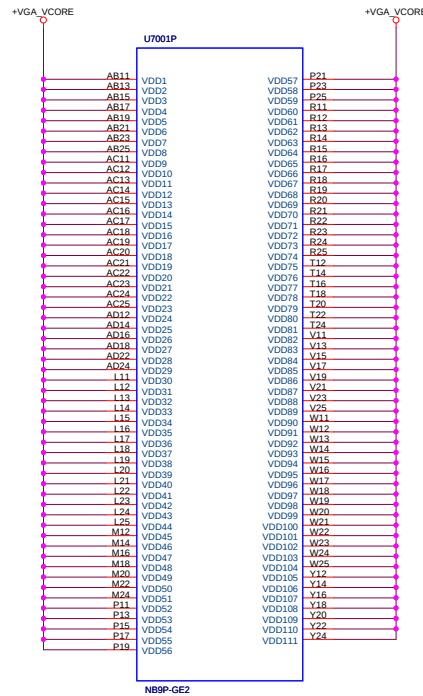
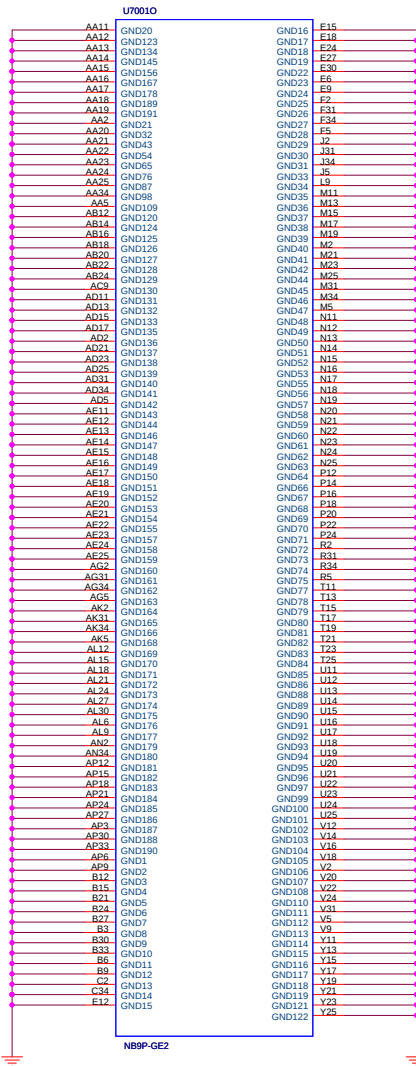


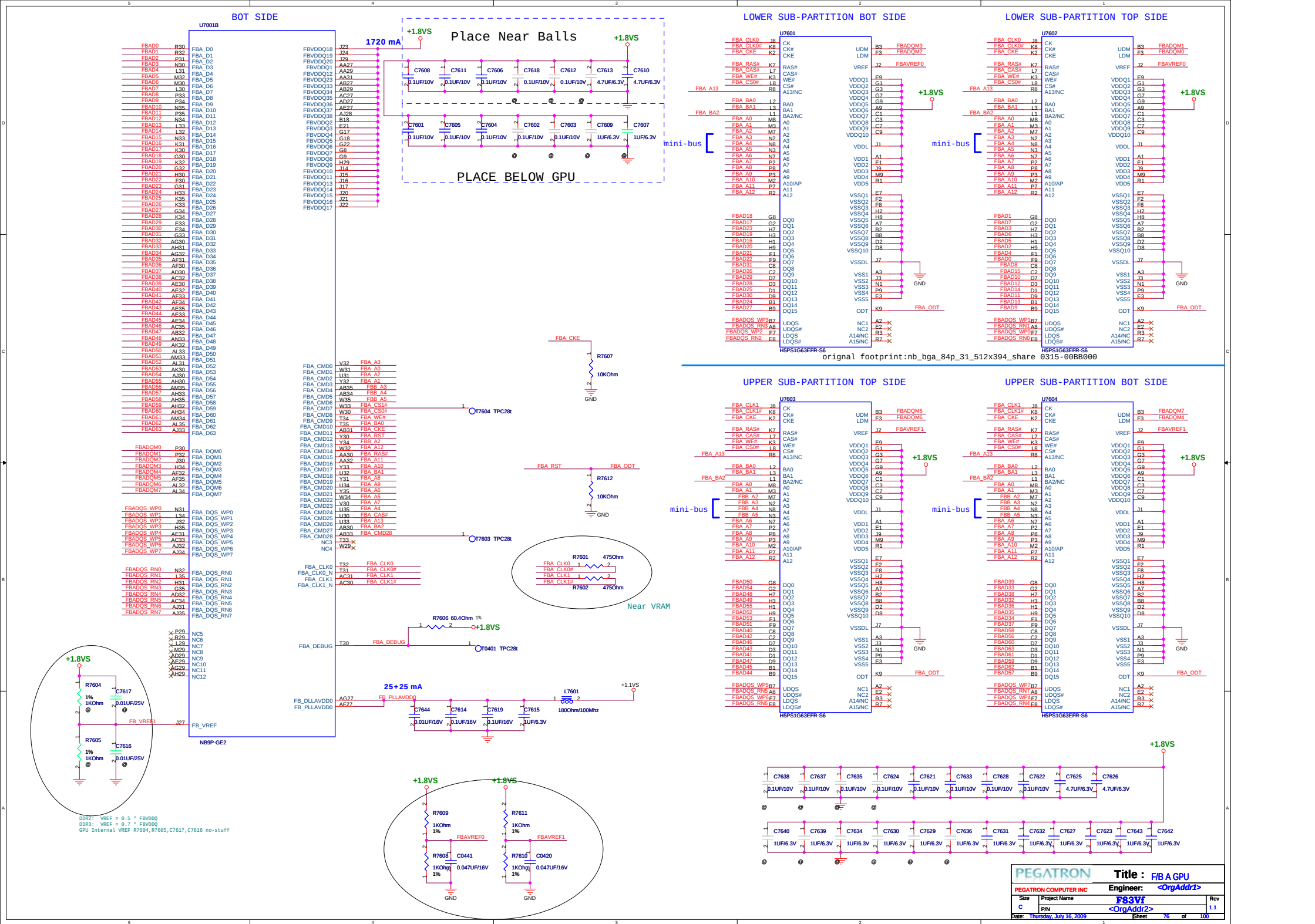
Default GPIO Assignment

GPIO0 -- AVAILABLE
 GPIO1 -- IFPC (HDMI) Hot Plug Detect
 GPIO2 -- Panel backlight brightness (PWM) (platform use EC PWM)
 GPIO3 -- Panel power enable
 GPIO4 -- Panel backlight ON/OFF
 GPIO5 -- GPU VID0
 GPIO6 -- GPU VID1
 GPIO7 -- GPU VID2 / FBVDD VID0
 GPIO8 -- OverTemp/GPU shutdown
 GPIO9 -- ThermAlert/Fan PWM
 GPIO10 -- FBVref Select
 GPIO11 -- SLI SYNC0
 GPIO12 -- AC power detect in
 GPIO13 -- PS_CONTROL0
 GPIO14 -- PS_CONTROL1
 GPIO15 -- IFPE Hotplug detect
 GPIO16 -- Dongle IFPC DVI mode
 GPIO17 -- Dongle IFPC HDMI mode
 GPIO18 -- Dongle IFPD DVI mode
 GPIO19 -- Dongle IFPD HDMI mode
 GPIO20 -- IFPD Hotplug detect
 GPIO21 -- NVGEM/ (IFPF Hotplug detect)
 GPIO22 -- SWAP Ready (SLI)
 GPIO23 -- AVAILABLE

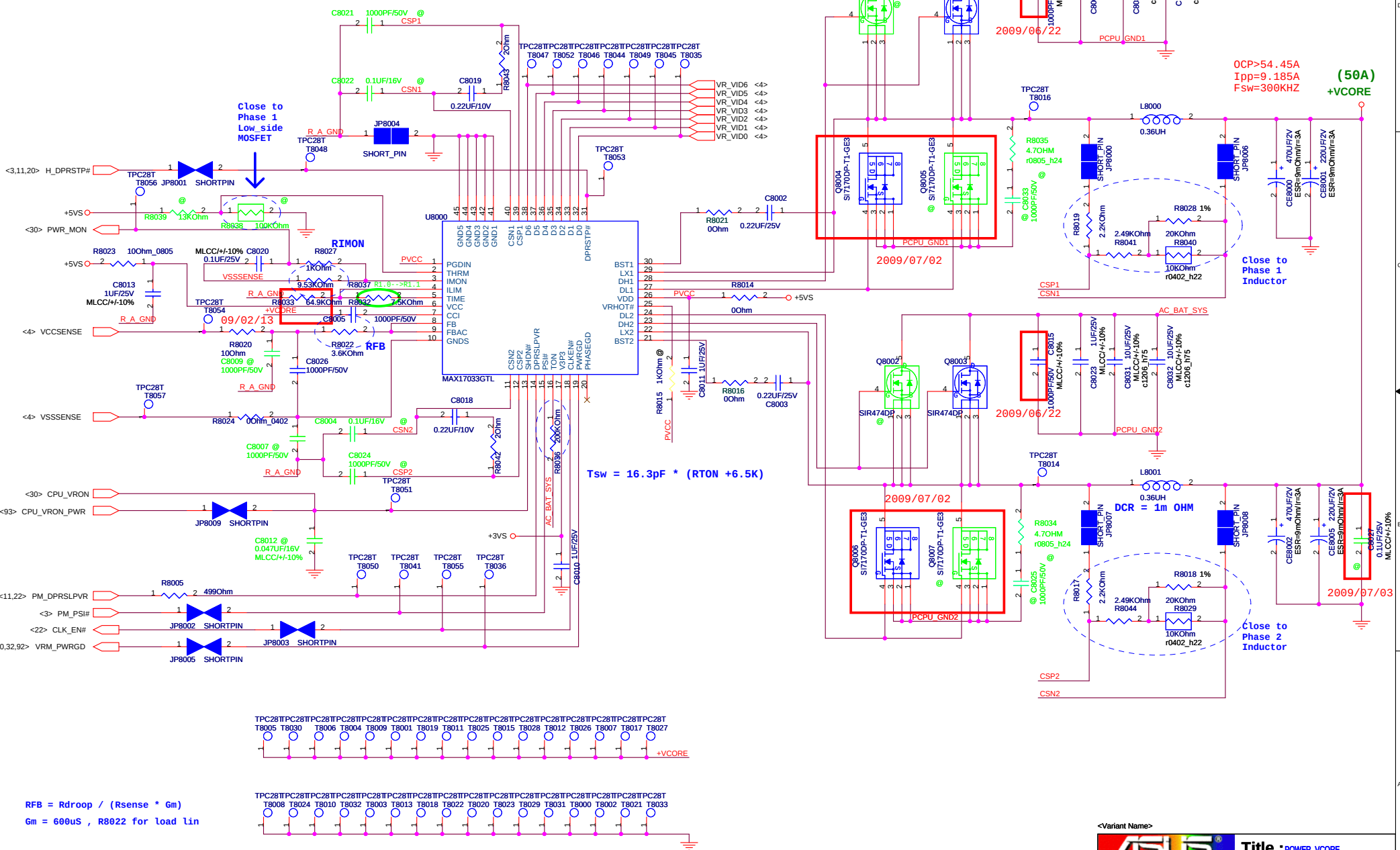
GPU_VID0	VID0	+VGA_VCORE
Low	0	0.9V
High	1	1.10V



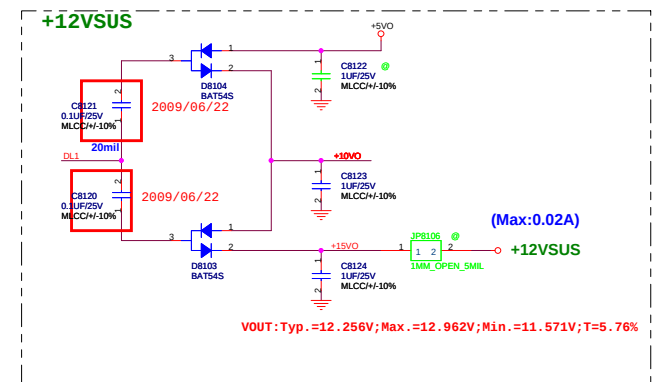




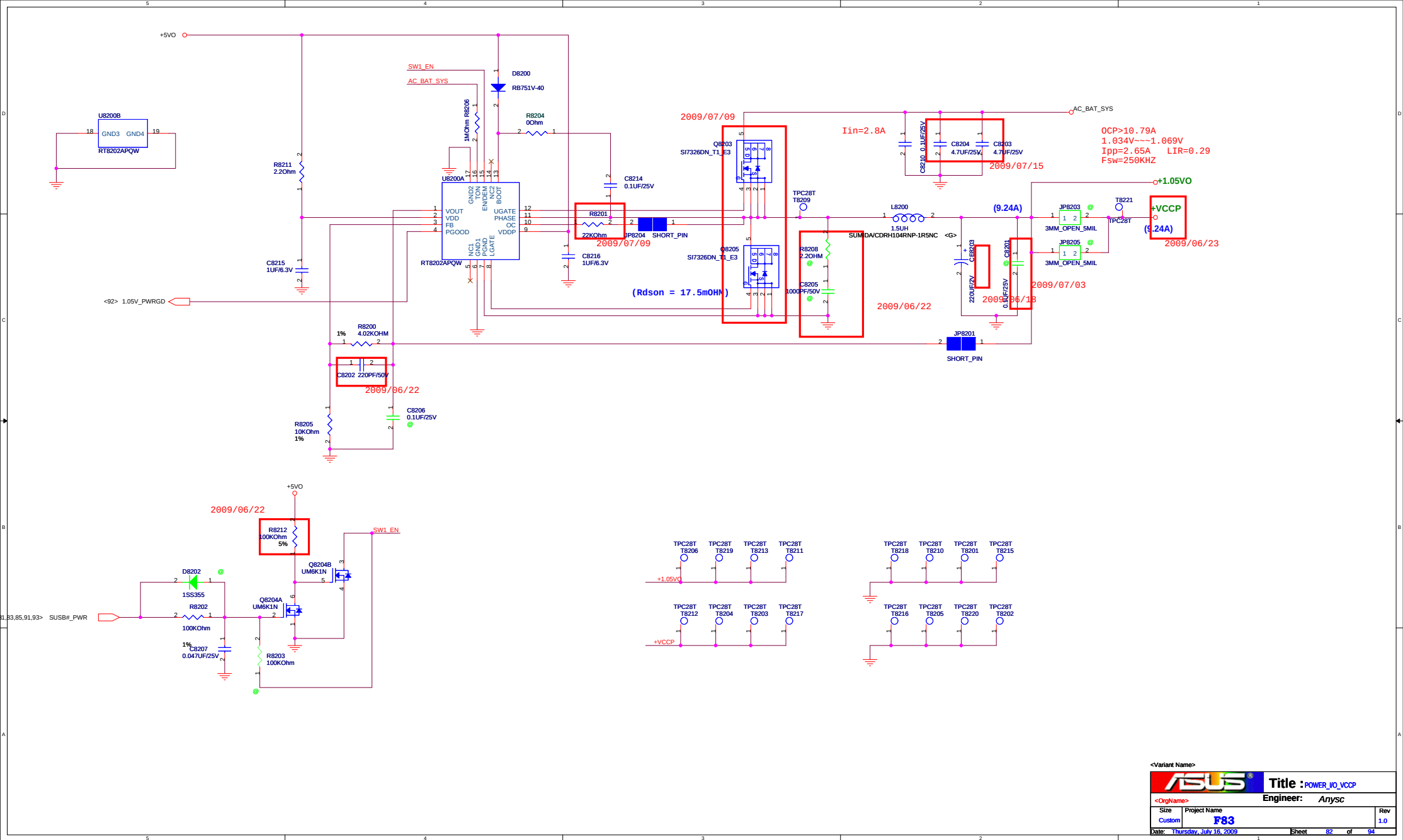
$IMON = Gm(IMON) * [(Vcsp1 - Vcsn1) + (Vcsp2 - Vcsn2)]$
 $RIMON = 0.9V / [IMAX * Rsense(MIN) * Gm(IMON_MIN)]$
 $Gm(IMON) = 2.4mS, Gm(IMON_MIN) = 2.36mS$

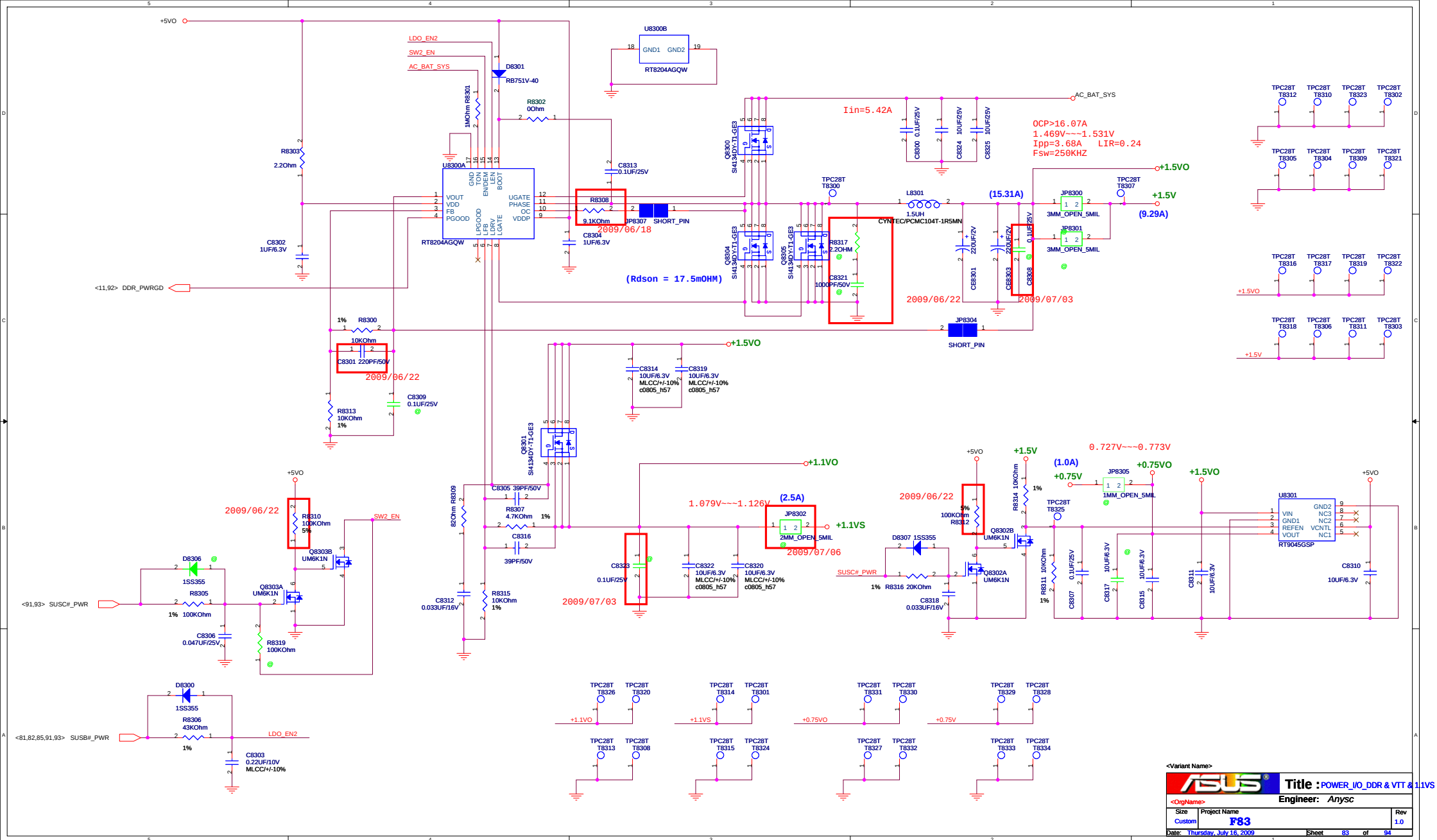


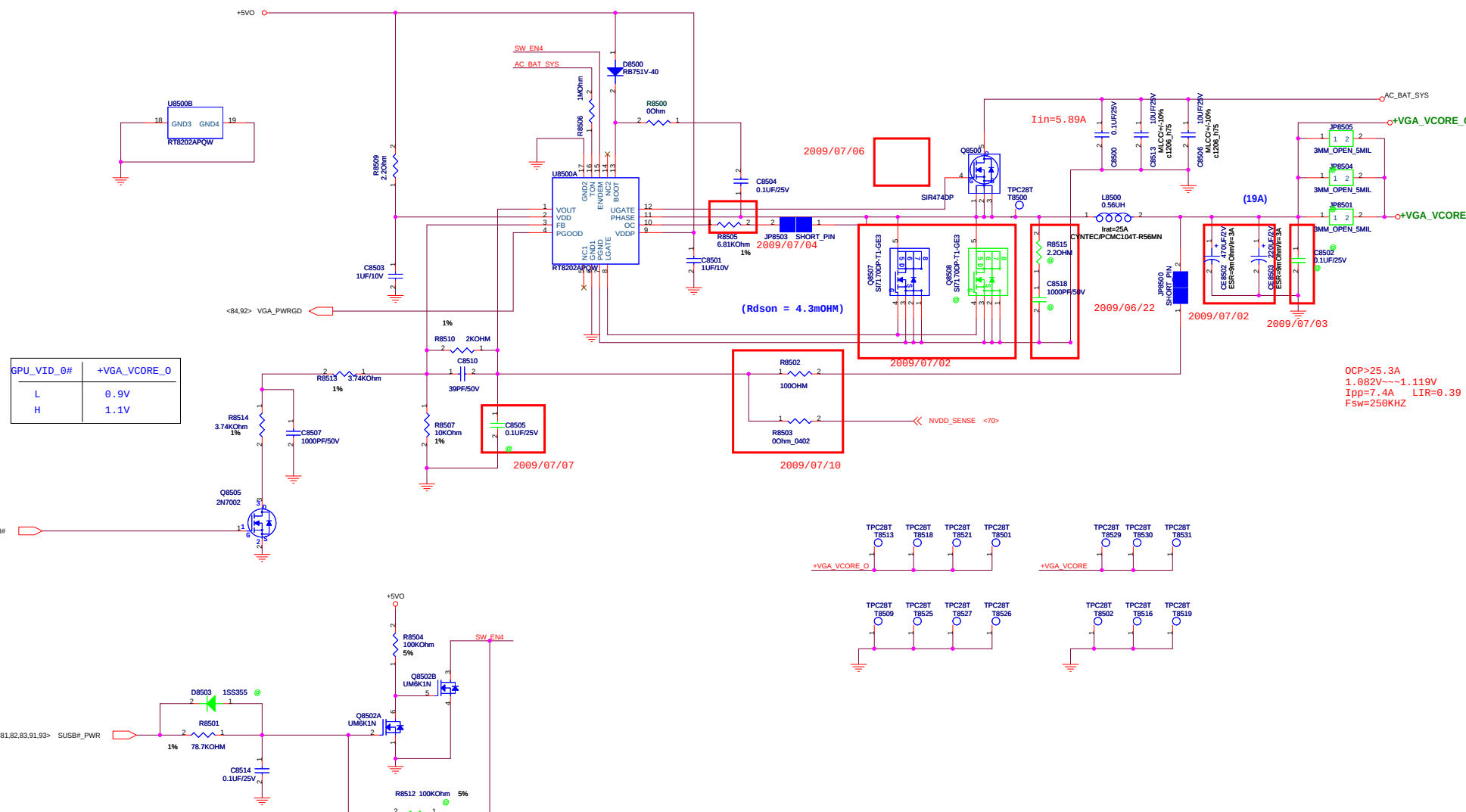
VENLDO:
Rising Edge:Max:2V;Typ:1.6V;Min:1.2V
Falling Edge:Max:1.06V;Typ:1V;Min:0.94V



<Variant Name>



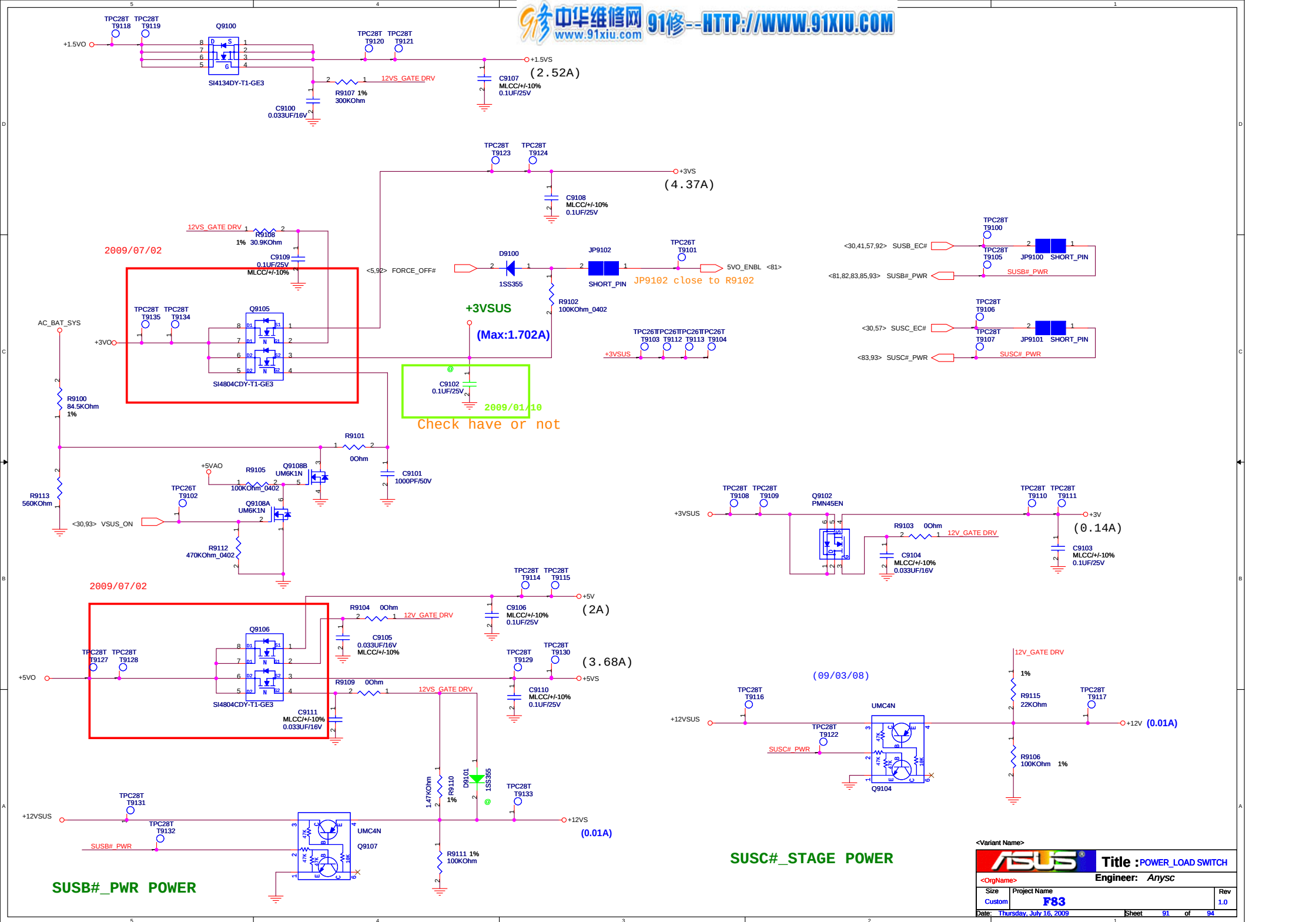




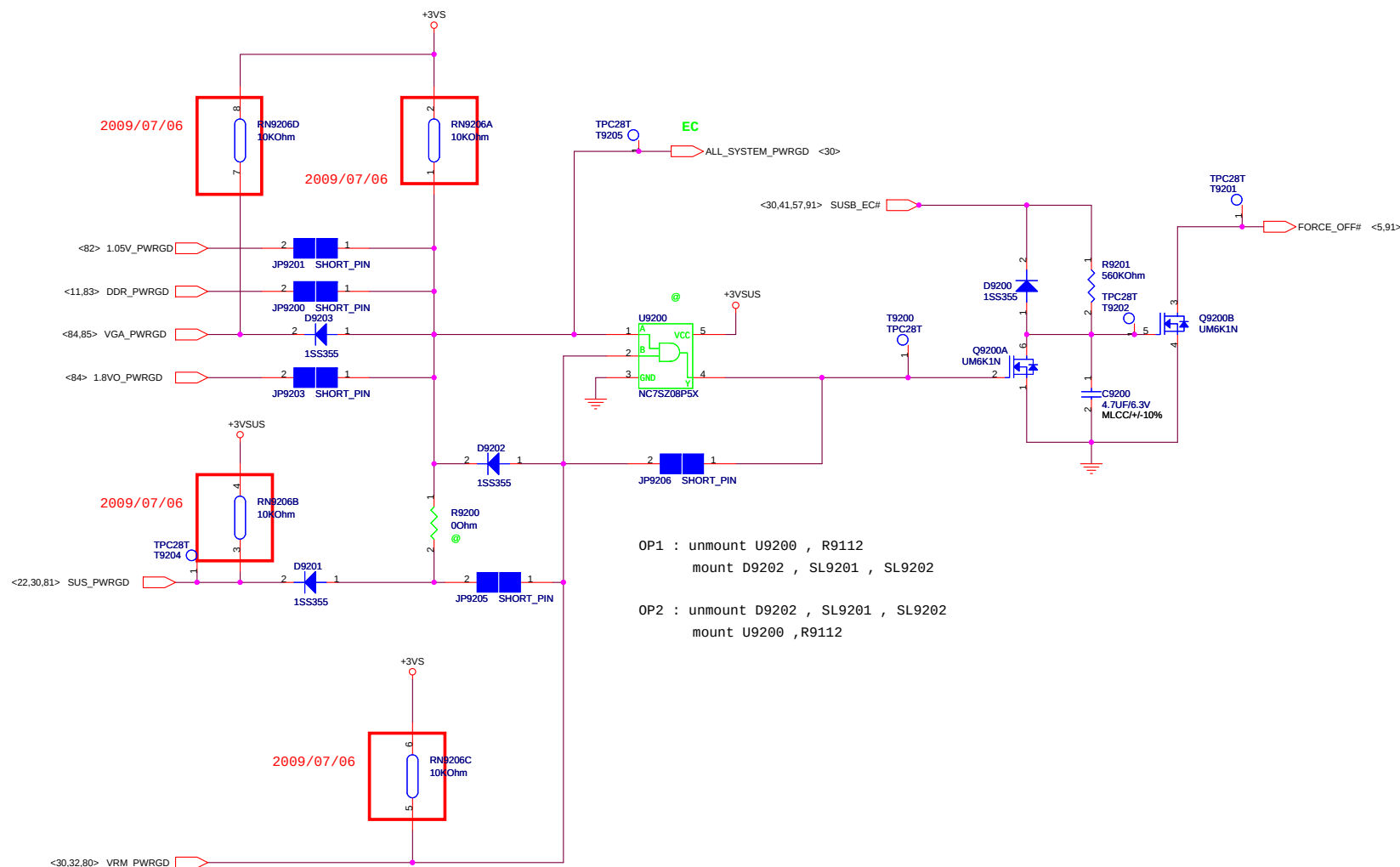
BATTERY IN DETECT

2009/07/08
delete batt detect





POWER GOOD DETECTOR



<Variant Name>

ASUS		Title : POWER_PROTECT	
<OrigName>		Engineer: AnySC	
Size	Project Name	Rev	
Custom	F83	1.0	
Date: Thursday, July 16, 2009	Sheet	92	of 94

FOR POWER TEST

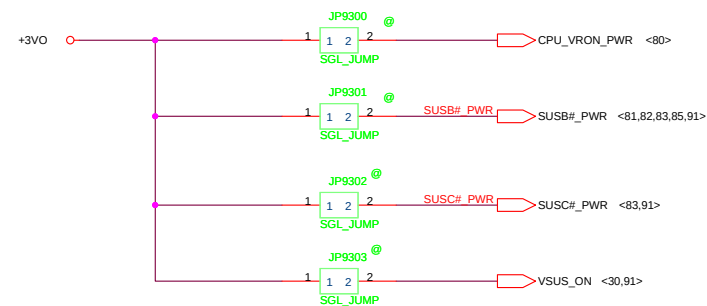
+3VA ○ → +3VA <20,22,30,45,57,60,81>
+3VSUS ○ → +3VSUS <20,21,22,23,24,30,33,37,45,53,56,91,92>
+5VSUS ○ → +5VSUS <23,56,81>
+12VSUS ○ → +12VSUS <24,81,91>

+5VO ○ → +5VO <81,82,83,84,85,91>
+3VO ○ → +3VO <81,91>
+1.05VO ○ → +1.05VO <82>
+1.5VO ○ → +1.5VO <83,91>
+0.75VO ○ → +0.75VO <83>
+1.8VO ○ → +1.8VO <84>
+1.1VO ○ → +1.1VO <83>
+VGA_VCORE_O ○ → +VGA_VCORE_O <85>

+3V ○ → +3V <21,33,41,44,53,57,61,63,91>
+5V ○ → +5V <9,31,44,45,52,57,91>
+12V ○ → +12V <37,91>
+1.5V ○ → +1.5V <7,8,9,11,13,14,57,83>

+3VS ○ → +3VS <3,7,8,11,14,15,22,23,24,29,30,32,36,40,41,44,45,46,48,50,51,53,56,57,70,72,74,80,91,92>
+12VS ○ → +12VS <24,40,45,48,57,91>
+5VS ○ → +5VS <23,31,36,46,48,50,51,56,57,80,91>
+1.8VS ○ → +1.8VS <57,73,76,77,84>
+1.5VS ○ → +1.5VS <4,14,23,41,53,57,91>
+VCCP ○ → +VCCP <3,4,5,10,11,13,14,20,23,29,57,82>
+0.75V ○ → +0.75V <7,8,11,83>
+1.1VS ○ → +1.1VS <57,70,72,73,76,77,83>
+VGA_VCORE ○ → +VGA_VCORE <75,85>
+VCORE ○ → +VCORE <4,5,80>

AC_BAT_SYS ○ → AC_BAT_SYS <45,80,81,82,83,84,85,88,91>
BAT ○ → BAT <88>
BAT_CON ○ → BAT_CON <60,88>



<Variant Name>

		Title : POWER_SIGNAL	
<OrgName>		Engineer: Anyisc	
Size Custom	Project Name F83		Rev 1.0
Date: Thursday, July 16, 2009		Sheet 93 of 94	

